Explainable and Fair Heart Disease Prediction: Understanding Bias and Fairness Trade-offs

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***Abstract* — Machine learning models trained on historical clinical records can carry forward the unequal care found in that data, and in heart disease this falls hardest on women, whose true cases are missed more often than men's. Two shortcomings run through the literature: the disparity is rarely measured, and explanation and fairness are treated as separate problems. This work treats them as one. It pursues two aims: to investigate where a sex-based performance disparity enters the model using explanation stratified by sex, without claiming to prove what causes it, and to develop an interactive dashboard that presents the fairness trade-offs a model faces under impossibility constraints. A single pipeline of three models, five fairness metrics, and three mitigation methods is applied to two datasets with sharply contrasting base rates: a 37.45 percentage-point sex gap in disease rate on one, and 0.75 on the other. The contrast lets the same pipeline show when fairness constraints bind and when they do not. On the wide-gap data the four decision-based metrics show substantial disparities together, sex is the feature the model leans on most for women while ranking low for men, and the false negative rate for women is roughly three times that for men; on the near-equal data the gaps and the reliance on sex all but vanish. Mitigation exposes the constraint: each method narrows one disparity while widening another, and reweighting breaks the model's calibration for women. The dashboard draws these trade-offs together, letting a user set a tolerance and see the resulting feasible region, so that fairness is presented for judgement rather than resolved in advance.**

Keywords — algorithmic fairness, explainable AI, SHAP, gender bias, heart disease prediction, impossibility result, base rate, fairness dashboard

I. Introduction

A. Background and motivation

Machine learning is increasingly used to support decisions in healthcare, from diagnosis to the allocation of resources. Because a model learns its patterns from historical records, where past care has been unequal between groups of patients, the model can carry that inequality forward instead of correcting it (Rajkomar et al., 2018). Whether such a model is fair therefore matters as much as whether it is accurate, and fairness cannot be taken for granted simply because a model performs well overall.

One such inequality is well documented in heart disease, which has long been regarded as a predominantly male condition. Yet women who develop it fare worse and receive recommended care less often than men, even when their symptoms are at least as severe (Bastian-Pétrel et al., 2024; Walsh, Jessup & Lindenfeld, 2019). This disadvantage is so long-standing that it carries its own name, the "Yentl syndrome", introduced by Healy in 1991 and persisting today (Bastian-Pétrel et al., 2024).

When clinical practice carries a systematic disadvantage of this kind, the data it generates carry it too, and a model trained on that data can learn the unequal care alongside the biology, not in place of it. The risk is not hypothetical: a commercial algorithm used in the United States scored Black patients as lower-risk than equally sick White patients. It did so because it used healthcare cost as a proxy for medical need, and less had historically been spent on the care of Black patients (Obermeyer et al., 2019).

Three terms are used throughout this work and are worth distinguishing at the outset: fairness, bias, and imbalance. Fairness, here, is a property of the decisions a model informs rather than of the model in isolation: whether those decisions favour or disadvantage a group defined by a sensitive attribute (Mehrabi et al., 2021). Bias is a systematic difference in the quality of those decisions for one group, one that clinical need does not explain. Imbalance is something narrower, the mere under-representation of a group in the data, which need not be unfair on its own. Each is defined more fully in the methodology; what matters here is that a disparity in the numbers is not yet evidence of a disadvantage.

Cardiovascular prediction is no exception. On a large clinical record, a recent study compared an established clinical score with several machine-learning models and found all of them biased against women. They gave women lower true positive rates and fewer positive predictions than men, a gender bias roughly seven times larger than any across racial groups (Li et al., 2023). The study could not identify where this bias entered the model, because it addressed the disparity only by adjusting the training data and never examined the features behind it. Two shortcomings run through this literature. The disparity is rarely measured in the first place. A systematic review of cardiac prediction models examined sixty of them in full and found only three that reported performance by sex, and none by race (Straw, Rees & Nachev, 2024). Nor has the turn to explainable methods closed the gap, since explanation and fairness are pursued as separate problems. Many heart-disease studies apply SHAP or LIME for interpretability, yet still treat "fairness" as class balance or reliability, or defer it altogether (Chakraborty et al., 2026; Rezk et al., 2024; Talukder et al., 2025). Explanation then serves to justify a prediction rather than to locate a disparity or weigh what reducing it would cost.

Explainability is often described as the bridge to clinical adoption: clinicians do not trust accuracy alone, but want to see which features drove a prediction, for the individual patient and across the group (Tonekaboni et al., 2019). Explanation and fairness meet at this point. When a model's errors fall unequally across groups, one possible explanation is that it relies on features differently across those groups. An explanation computed across all patients cannot reveal this, because it averages the sexes together. Separating the explanation by sex is therefore what makes the difference visible, and it is this step that relates a measured disparity to the features associated with it.

A further constraint shapes what any such work can achieve. When the rate of disease differs between two groups,

no model can satisfy every definition of fairness at once: the definitions conflict, and improving one is paid for in another (Kleinberg, Mullainathan & Raghavan, 2017; Chouldechova, 2017). This is a mathematical result, not a shortcoming of any particular model, and it is set out in detail in the related work. Its consequence for the present work is that fairness cannot be solved, only presented, so that the choice of what to trade is left exposed rather than settled in advance.

B. Aims and objectives

This project treats explanation and fairness as one problem rather than two, addressing the separation described above. It pursues two aims. The first is to investigate where any sex-based performance disparity enters the model: explanation stratified by the sensitive attribute is used to ask which features carry that disparity, not to prove what causes it. The second is to develop an interactive dashboard that presents the fairness trade-offs a model faces under these impossibility constraints. The dashboard is presented here as an educational prototype that makes these trade-offs visible; with further development, it could become a standard tool within model-development workflows. Sex is the sensitive attribute throughout, but it serves as a worked example: the method depends only on a group label, and the same pipeline would apply to another attribute such as ethnicity.

Four objectives carry these aims. The first is to show how a sex-based performance disparity can be measured across five fairness metrics, so that the conditions under which those metrics conflict can be observed rather than assumed, with two datasets of contrasting base rates as a case study. The second is to show how a model's reliance on features differs between male and female patients, so that a measured disparity can be investigated in relation to the features associated with it rather than left unexplained. The third is to examine how bias mitigation shifts the trade-offs between metrics, so that the cost of reducing one disparity can be weighed against the others rather than hidden. The fourth is to draw these trade-offs together in a dashboard on which a user sets an acceptable tolerance, so that the choice of what to trade is made openly rather than fixed in advance. The contribution is to make the fairness trade-off visible and adjustable: two datasets of contrasting base rates, explanation stratified by sex, and a tolerance-based dashboard are brought together in a single pipeline that presents these trade-offs rather than resolving them in advance.

II. Related work

A. Fairness in healthcare ML

Fairness in machine learning is commonly framed as the absence of systematic disadvantage toward a group defined by a sensitive attribute, inherited from the data rather than invented by the model (Mehrabi et al., 2021). In clinical diagnostic tasks a model is judged on how equally it performs across groups rather than how it allocates a resource, so the natural metrics concern equality of error rates (Feng et al., 2025). A recent review of structured health data reports that group-level metrics of this kind dominate the literature, an approach the present work follows (Ruback et al., 2026).

What counts as a systematic disadvantage, however, is not settled by a single measure. It can mean that a group receives fewer positive predictions than another (Feldman et al., 2015). It can mean that the model misses a larger share of that group's genuine cases, or raises more false alarms for it (Hardt, Price & Srebro, 2016). Or it can mean that a positive prediction is less trustworthy for one group than another (Chouldechova, 2017). These readings need not agree, and a model fair on one can be unfair on another. Which of them can hold together at once is the subject of the next section.

B. The impossibility result

The tension that motivates this work is mathematical, not a matter of engineering effort. When outcome prevalence differs between two groups, a classifier cannot hold its positive predictive value equal across them and also equalise its false positive and false negative rates. Once prevalence is fixed, these three quantities are locked together by a single relation (Chouldechova, 2017):

$$FPR = \left(\frac{p}{1-p}\right) \times \left(\frac{1-PPV}{PPV}\right) \times (1-FNR) \quad (1)$$

where p is the prevalence within a group, PPV the positive predictive value, and FPR and FNR the false positive and false negative rates. With PPV held equal across groups, a difference in p forces FPR and FNR to differ, since the relation cannot otherwise hold on both sides.

The result was derived for a recidivism tool, but it depends on prevalence alone rather than on the setting (Chouldechova, 2017). Calibration gives a second route to the same conclusion: within-group calibration and balanced error across groups cannot both hold unless prediction is perfect or the groups share a base rate (Kleinberg, Mullainathan & Raghavan, 2017).

These results assume exact equality of the fairness metrics across the groups, a difference of exactly zero. Relaxing that assumption softens them. Allowing a small margin of tolerance turns the single feasible point into a whole region of workable models, and that region grows as the base-rate gap narrows and the classifier improves (Bell et al., 2023). This is what a trade-off dashboard depends on. It turns tolerance into a parameter the user sets, instead of a fixed impossibility. Whether the impossibility applies depends on how the data is read: if the base-rate difference is taken as real it holds, and if taken as an artefact of biased recording it no longer applies (Friedler, Scheidegger & Venkatasubramanian, 2021). This work takes the difference as real, since the model is trained on the rates as recorded. That reading is supported by descriptive checks on the UCI heart disease data used in this study: the gap remains within Cleveland and Hungary, the two hospitals with the largest female samples, while the mean ages of women and men differ by only about a year, as given in Appendix C. These checks are descriptive rather than causal, and cannot settle whether the recorded rates reflect a genuine difference in prevalence, the historical under-diagnosis of women, or both, a question the discussion returns to.

C. Explainability methods

Explaining an individual prediction is widely presented as a precondition for clinical trust in a complex classifier (Tonekaboni et al., 2019). The dominant tool is SHAP, which gives each feature a contribution by treating the features as players in a cooperative game (Lundberg & Lee, 2017). These values are the only attribution satisfying local accuracy, consistency and missingness at once, placing explanation on firmer footing than earlier heuristics (Lundberg & Lee, 2017). An older alternative, LIME, fits a linear model around one prediction and works on any classifier, but samples at random, so repeated runs can disagree (Ribeiro, Singh & Guestrin,

2016). SHAP is therefore preferred here as the primary method, LIME as a check.

For tree ensembles, an exact form of SHAP can be computed in polynomial time, making it practical to explain every patient (Lundberg et al., 2020). That work is also a close precedent, using interaction values to show the male-female mortality risk difference shifting across the lifespan, the kind of sex-stratified reading the present work builds on (Lundberg et al., 2020). Explanation is therefore well developed for describing a prediction, but rarely turned towards whether a model's errors fall unevenly across groups.

D. Gender bias in cardiac prediction

A few studies have examined this error-level disparity directly in cardiac prediction. A systematic review that examined sixty cardiac models and reproduced their reported algorithms recorded a false negative rate significantly higher for women in thirteen of sixteen experiments and a higher false positive rate for men (Straw, Rees & Nachev, 2024). Disease was thus under-predicted for women and over-predicted for men, and a range of remediation techniques failed to remove the gap, sometimes reversing it with the dataset (Straw, Rees & Nachev, 2024). A study of a large clinical record found the bias against women far larger than any across racial groups. The equal-opportunity difference reached 0.131 to 0.136 by sex against 0.018 or less by race, expressed as a lower true positive rate for women (Li et al., 2023). It addressed the disparity only by adjusting the training data, without examining the features behind it (Li et al., 2023).

Closer to the present datasets, sex-specific models on the cardiovascular record used here raised recall for women but measured only predictive performance, not fairness or its trade-offs (Roy, Dubey & Roy, 2026). A diabetic-patient framework cast model selection as an accuracy-fairness trade-off and found women assigned lower risk, but did not engage the impossibility results (Yang et al., 2025). An electrocardiogram-model audit invoked those results directly and found that preprocessing left the disparity intact while group-specific thresholds reduced it (Kaur et al., 2024). Reporting several metrics at once can also send conflicting signals, one sex favoured on discrimination and the other on calibration (Khan et al., 2025). The disparity is well established, yet explanation is seldom tied to fairness, trade-offs are rarely presented under the impossibility constraints, and a single dataset leaves unclear when they become binding.

E. Bias mitigation techniques

Once a disparity has been measured, the ways of reducing it differ mainly in where they intervene in the pipeline. Some act on the data before any model is trained. Oversampling the minority class is one route: it interpolates synthetic points between a sample and its neighbours, and so widens the decision region instead of merely duplicating it as replication would (Chawla et al., 2002). Reweighting is another: it assigns a weight to each group-class combination to balance the training data, and changes no instance's label in doing so (Kamiran & Calders, 2012). Other methods wait until the model is fitted, then adjust its score with a separate threshold for each group to equalise error rates (Hardt, Price & Srebro, 2016). This can narrow the gap in error rates without forcing both groups to be flagged as having disease at the same overall rate. Toolkits such as Fairlearn now gather these metrics and mitigators behind one interface (Bird et al., 2020). None of this comes for free. A reduced disparity is paid for in accuracy, and the exchange is often close to linear (Kamiran & Calders, 2012). In the end there is no single right fix: each choice gives a different balance, one to be shown to the user rather than fixed in advance.

F. Fairness dashboards

Presenting trade-offs rather than resolving them calls for an interface that can carry them, and a line of interactive fairness tools has begun to do so. One such tool lets a user probe a model without any access to its internals: a slider sets a separate threshold for each group, and the error rates update in place as it moves (Wexler et al., 2020). A second lays several fairness metrics side by side across intersectional subgroups, each group's base rate shown alongside its scores (Cabrera et al., 2019). Tools of this kind make a disparity visible, yet stop short of tying it back to the features on which the model relies, since explanation by attribution is not part of them. The step still missing is to bring these strands together, so that a dashboard rests on the impossibility constraints and links the observed disparity to how the model uses features differently across groups.

Reviews of this literature note that the accuracy-fairness trade-off frequently goes unreported and that one metric alone can mislead, since fairness on one is paid for on another (Ruback et al., 2026). Explanation is not integrated into any of this. The present work takes up these threads together.

III. Methodology

The methodology is organised as a single pipeline applied identically to both datasets, so that their results can be compared on equal terms. This section first defines the terms on which fairness is judged, then works through the pipeline shown in Fig. 1: data preprocessing, model training, explanation, fairness measurement, bias mitigation, and finally the interactive dashboard.

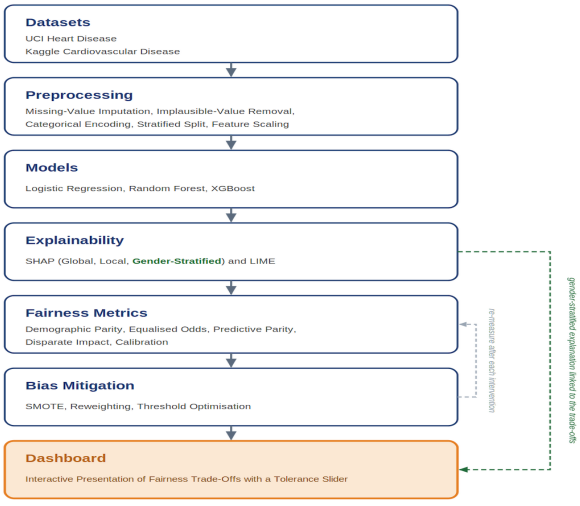


Fig. 1. Overview of the methodology pipeline, applied identically to the two datasets.

A. Operational Definitions and Terminology

Fairness, in this work, is a property of decisions. A model counts as fair when the decisions taken on the strength of its predictions do not lean systematically for or against a group defined by a sensitive attribute (Li et al., 2023; Mehrabi et al., 2021). Five metrics put this into practice, each expressing a distinct and at times competing sense of what an even outcome means. None of them can settle the question alone. In keeping with the group-level convention common to structured health

data, the metrics compare whole groups rather than individuals (Ruback et al., 2026).

Throughout, bias is kept separate from imbalance. Imbalance is the numerical under-representation of one group in a dataset. On its own it is not unfairness, since a sample with fewer women than men may simply reflect the population it was drawn from. Bias, by contrast, is a systematic difference in the quality of the decisions a model makes for one group, one that clinical need does not explain. It shows itself most clearly as a higher false negative rate for one sex, meaning that a larger proportion of that sex's genuine disease cases go undetected (Li et al., 2023). This distinction decides how the position of women is read here: their forming a minority is, on its own, only an imbalance, and becomes a disadvantage only if the model misses genuinely ill women more often than genuinely ill men. Which of the two is at work cannot be settled from the data alone, so the aim is to investigate that question rather than resolve it.

This work focuses on sex because the disparity there is both documented and comparatively under-examined in machine-learning models. Throughout, the sensitive attribute is the biological sex recorded in the data, and the terms "gender" and "gender-stratified" refer to this same variable.

The competing senses of fairness cannot generally be reconciled. As set out in the related work, when disease prevalence differs between two groups no classifier can equalise every metric at once (Chouldechova, 2017; Kleinberg, Mullainathan & Raghavan, 2017). This is what makes the two datasets used here, with their contrasting sex-based disease-rate gaps, a useful test of when the constraints become binding.

B. Datasets

The study is built on two datasets rather than one, because a single dataset cannot separate a group's base-rate effect on fairness from the other ways datasets differ. Two datasets with sharply contrasting sex-based disease-rate gaps give two cases, one where the impossibility constraints bind sharply and one where their effect is minimal, both open and suited to a reproducible comparison. The first is the UCI heart disease dataset compiled from four cardiology centres and distributed through Kaggle (Sony, 2020). It contains 920 patients, described by ten clinical features retained for modelling after preprocessing, and its patients are predominantly male, with women making up only 21.1 per cent of the sample. The disease rate differs sharply between the sexes: 63.22 per cent of male patients are recorded as having heart disease, against 25.77 per cent of female patients, a gap of 37.45 percentage points. The dataset also carries a quiet data-quality problem in that a group of patients have a cholesterol reading stored as zero, which is clinically impossible and is treated as a hidden missing value.

The second is the Kaggle cardiovascular disease dataset of routine examination records (Ulianova, 2019). After the removal of biologically implausible measurements, it contains 68,653 patients described by eleven features, and its composition is close to the reverse of the first: women make up 65.1 per cent of the sample. Here the disease rate is almost identical across the sexes, at 49.21 per cent for women and 49.96 per cent for men, a gap of only 0.75 percentage points. The two datasets also differ in source and feature composition, not only in balance.

TABLE 1. COMPARISON OF THE TWO DATASETS AFTER PREPROCESSING

Property	UCI Heart Disease	Kaggle Cardiovascular Disease
Patients (n)	920	68,653
Features	10	11
Female / Male (%)	21.1 / 78.9	65.1 / 34.9
Disease rate, female (%)	25.77	49.21
Disease rate, male (%)	63.22	49.96
Base-rate gap (pts)	37.45	0.75

The contrast in the final row of Table 1 is the empirical centre of this work. The large gap in the UCI heart disease data should not be read on its own as evidence of bias, since a difference of this kind may instead reflect a genuine difference in disease prevalence between the sexes, and the two cannot be told apart from the data alone (Bastian-Pétrel et al., 2024).

C. Preprocessing

The UCI dataset required the most cleaning. Three columns, ca, thal and slope, were dropped, because they were missing for 66.41, 52.83 and 33.59 per cent of patients respectively, and imputing such a large share would have introduced too many artificial values. The zero cholesterol readings flagged earlier, together with a single zero blood-pressure reading, are clinically impossible and were treated as missing. The remaining numerical gaps were filled with the column median, which is more robust than the mean to the skew in these measurements, and the categorical gaps with the most frequent value. The categorical clinical features were then integer-encoded, and sex was encoded with women as 0 and men as 1.

The Kaggle dataset needed no imputation, since it contained no missing values, but it carried a different problem in the form of biologically implausible measurements. Rows were removed where a reading was physically impossible in a living adult. This removed 1,347 of the 70,000 rows, most of them (1,234) with a systolic reading below their diastolic one, and left the 68,653 patients described earlier. Age, recorded in days, was converted to years, and gender was encoded with women as 0 and men as 1.

From this point the two pipelines were identical. Each dataset was split into training and test sets in an 80 to 20 ratio, stratified by the target so that the disease rate was preserved in both sets, with a fixed random seed for reproducibility (Pedregosa et al., 2011). The features were then scaled to a common 0 to 1 range using a MinMax scaler fitted on the training set alone and applied to the test set, so that no information from the test set entered the scaling.

D. Models

Three models were trained and compared, chosen to represent three different model families: logistic regression as a linear baseline, random forest, and XGBoost. Logistic regression fits a linear decision boundary and serves as a transparent reference point against which the two ensembles are read. Random forest grows many decision trees, each on a bootstrap sample and a random subset of features, then combines their votes, so that adding more trees does not raise the generalisation error (Breiman, 2001). XGBoost instead builds its trees in sequence, each correcting the errors of the last, with a regularised objective that penalises complexity to control overfitting (Chen & Guestrin, 2016). Each model's

hyperparameters were tuned with ten-fold cross-validation on the training set, using accuracy as the selection criterion, so that the test set was held back for final evaluation alone.

E. Explainability

The trained models produce predictions but do not show which features drive them, so SHAP is applied to both tree ensembles, while LIME provides a local cross-check on the tuned XGBoost (Lundberg & Lee, 2017; Ribeiro, Singh & Guestrin, 2016). SHAP is used in three separate roles, kept apart throughout: a global ranking of which features matter across all patients, a local explanation of a single patient's prediction, and a comparison of feature importance computed for women and men separately. This third role is the one that links explanation to fairness, since a model that relies on different features for each sex can produce an unequal error rate. Because the two ensembles are tree-based, their SHAP values are computed exactly with a tree explainer, which is fast enough to cover every patient in the test set individually (Lundberg et al., 2020).

F. Fairness metrics

Whether the model produces unequal outcomes for female and male patients is assessed using five fairness metrics. Four are based on the model's decisions, are computed with Fairlearn, and reduce to a single summary value: either a difference or a ratio (Bird et al., 2020). The fifth, calibration, is based on the model's probability scores and is assessed separately. Throughout, $\hat{Y}$ denotes the model's prediction and Y the true outcome, each equal to 1 when disease is present, and A the sex, with values f for female and m for male.

Demographic parity holds when the model predicts disease at the same rate for each sex (Hardt, Price & Srebro, 2016):

$$P(\hat{Y} = 1 \mid A = f) = P(\hat{Y} = 1 \mid A = m) \quad (2)$$

where the gap is the absolute difference between these two rates. Equalised odds requires the true and false positive rates to match across the sexes (Hardt, Price & Srebro, 2016):

$$P(\hat{Y} = 1 \mid Y = y, A = f) = P(\hat{Y} = 1 \mid Y = y, A = m), y \in \{0,1\} \quad (3)$$

reported as the larger of the absolute true positive rate and false positive rate gaps. Predictive parity requires a positive prediction to be equally trustworthy for each sex, that is, to have equal precision (Chouldechova, 2017):

$$P(Y = 1 \mid \hat{Y} = 1, A = f) = P(Y = 1 \mid \hat{Y} = 1, A = m) \quad (4)$$

measured as the absolute difference in precision between the groups. Disparate impact compares the rate at which each group receives a positive prediction, expressed as a ratio that falls below one when the rates differ (Feldman et al., 2015):

$$\frac{min(P(\hat{Y} = 1 \mid A = f), P(\hat{Y} = 1 \mid A = m))}{max(P(\hat{Y} = 1 \mid A = f), P(\hat{Y} = 1 \mid A = m))} \geq \tau, \tau = 0.8 \quad (5)$$

where a ratio of 1 indicates equal positive-prediction rates and $\tau = 0.8$ is the four-fifths threshold below which disparate impact is flagged (Feldman et al., 2015). The ratio is computed symmetrically as the smaller rate over the larger, using the demographic parity ratio provided by Fairlearn (Bird et al., 2020). Demographic parity and disparate impact are two readings of the same comparison, one subtracting the two flagging rates and the other dividing them, so only three of the four carry separate information. Unless otherwise stated, they are calculated on the held-out test set at a decision threshold of 0.50.

The fifth metric, calibration, works on the probability scores rather than the decisions, so it does not reduce to a single gap. A model is calibrated when a predicted risk carries the same meaning for each sex: among patients given a risk of around 0.7, close to 70 per cent truly have disease, and this should hold for both groups (Van Calster et al., 2019). Calibration is therefore read from reliability curves compared for female and male patients. Because it depends on the scores alone, it is unaffected by the decision threshold, unlike the four metrics above. The group prevalences here differ, so these metrics cannot all hold at once; which trade-offs actually appear is shown in the results.

G. Bias mitigation

Three mitigation methods are applied, each intervening at a different stage of the pipeline: two before the model is fitted and one after. All three are demonstrated on the tuned XGBoost, the main model of this study; whether the same shifts appear in the other model families is left outside the scope of this study, as are in-processing methods that change the learning algorithm itself. SMOTE-NC, the variant suited to a mix of continuous and categorical features, oversamples the minority class, drawing on five nearest neighbours to generate each synthetic patient (Chawla et al., 2002). Reweighting takes a different route: each sex-and-outcome group is given the rate expected under independence divided by the rate observed, weakening the association between sex and the outcome without changing any label (Kamiran & Calders, 2012). The last method leaves the trained model untouched, setting a separate decision threshold for each sex, so that the predicted probabilities, and with them the explanations and the calibration, stay as they were (Hardt, Price & Srebro, 2016; Bird et al., 2020). Each is applied only to the training data, after the split, so that the test set still reflects the real population. The same data and the same model thus yield different trade-offs depending on the intervention: there is no single correct model here, only choices.

H. Dashboard design

The dashboard is written for the person who trains a clinical prediction model and has to judge, before it reaches a clinician, whether its behaviour is acceptable. It does not choose a fair model for that person. Instead, it leaves the decision to them: the user sets how large a difference between the two patient groups they are willing to accept, and the dashboard reports whether the model can meet it. This is the deliberate centre of the design. The impossibility result says that no setting satisfies every fairness metric at once when the two groups differ in their underlying rates of disease, so the dashboard does not try to resolve that conflict. It presents it, turning the abstract mathematics of the earlier sections into a question the user can see and answer.

The dashboard applies this choice through its tolerance setting: for the three difference metrics a value passes when it sits no higher than the tolerance, and for disparate impact the ratio must reach at least one minus the tolerance, with 0.8 kept as a fixed four-fifths reference. The metrics reported in the results are calculated at a single decision threshold of 0.50. The dashboard keeps this as the starting point but lets the user change it, setting a separate threshold for female and for male patients and seeing the metrics respond, in the manner of an interactive tool that probes a model through group-specific thresholds without access to its internals (Wexler et al., 2020). For the probability-producing methods it works through every pair of group-specific thresholds and shades those where all

four decision-based metrics fall within the chosen tolerance; the threshold optimiser is reported on its own, since it fixes a decision rather than leaving a pair for the user to search. This shaded set is a feasible region: loosening the tolerance widens it, tightening it shrinks the region, and where the two groups differ sharply in their base rates it can disappear altogether (Bell et al., 2023). Whether that set is empty is itself the answer, shown at once rather than searched for.

Throughout, the metrics are placed side by side as a bar chart rather than reduced to a single figure, so that the trade-off stays in view. Each is reported both as a value and as a count of patients: a fairness gap reads as a decimal against the tolerance and as so many women flagged in every hundred against so many men, and the errors behind it are drawn as a hundred squares, one to a patient, with the individual missed cases and false alarms listed beneath. A second view collects the supporting analysis behind the trade-off, holding the gender-stratified explanation, the calibration curves, and a live comparison of the mitigation methods in one place. The dashboard makes fairness a judgement, not a calculation.

IV. Results

A. *Model performance*

On the UCI data the two tree-based models overfitted at their default settings, reaching a perfect training accuracy of 1.000 against a lower test accuracy, while the logistic regression baseline did not. Tuning brought this under control, so that the comparison in Table 2 rests on how well the models generalise. After tuning, XGBoost was the strongest on the UCI data (test accuracy 0.8152, AUC-ROC 0.9082), the random forest close behind at 0.9033. On the Kaggle data the models sat in a narrower and lower band, the tuned XGBoost reaching a test accuracy of 0.7288 and an AUC-ROC of 0.7991, which reflects how much less separable the two classes are. XGBoost was carried forward as the main model on both datasets, with the other two kept for comparison.

TABLE 2. Model performance on the two datasets after tuning

Dataset and model	Training accuracy	Test accuracy	AUC-ROC
UCI, logistic regression	0.7853	0.7663	0.8772
UCI, random forest	0.8546	0.7935	0.9033
UCI, XGBoost	0.8152	0.8152	0.9082
Kaggle, logistic regression	0.7290	0.7212	0.7903
Kaggle, random forest	0.7735	0.7294	0.7979
Kaggle, XGBoost	0.7402	0.7288	0.7991

B. *Fairness under a wide base-rate gap*

The wide base-rate gap in the UCI data places it where sharp trade-offs are expected, and this is what the fairness metrics show. Measured on the tuned XGBoost, all four decision-based metrics show substantial disparities at the default threshold (Table 3). Of these the disparate impact ratio is the most striking, since at 0.3406 it falls far below the 0.8 threshold of the four-fifths rule, so women are flagged as having disease at only about a third of the rate of men. That these gaps appear together is no coincidence: the unequal base rates make predictive parity and equalised error rates impossible to hold at once, though the size of each gap also depends on the model and the decision threshold. This is the impossibility result seen in the data rather than in theory.

TABLE 3. Fairness metrics for the tuned XGBoost on the two datasets

Fairness Metric	UCI Heart Disease	Kaggle Cardiovascular Disease
Demographic parity difference	0.4426	0.0094
Equalised odds difference	0.2580	0.0024
Predictive parity difference	0.2050	0.0110
Disparate impact ratio	0.3406	0.9792
False negative rate, female	0.3750	0.3226
False negative rate, male	0.1170	0.3203

This pattern recurs across the three model families rather than resting on one alone. The disparate impact ratio stays well below 0.8 whether the model is a linear baseline, a bagged ensemble or a boosting method (0.2129, 0.4093 and 0.3406), though the size of the other gaps varies by model (Table 4). The substantial disparities are not specific to any single algorithm.

TABLE 4. Fairness metrics across the three models on the UCI data

Fairness Metric	Logistic regression	Random forest	XGBoost
Demographic parity difference	0.5283	0.4123	0.4426
Equalised odds difference	0.3630	0.2337	0.2580
Predictive parity difference	0.0200	0.1981	0.2050
Disparate impact ratio	0.2129	0.4093	0.3406

The model behaviour behind the disparity is read from the gender-stratified SHAP values, keeping the ranking and the signed direction apart. For female patients sex is the single most important feature, ahead of chest pain, while for male patients it drops to fifth place behind the clinical measurements (Fig. 2). Across all patients sex ranks only fourth, so separating the two groups is what makes the difference visible; the global and local SHAP plots, with a LIME cross-check on the direction, are given in Appendix B. The direction confirms the split: the mean SHAP value for sex is -0.6941 for women and +0.1909 for men, pushing female predictions towards no disease and male ones towards disease, an effect about three and a half times stronger for women. The false negative rate is 0.3750 for female patients against 0.1170 for male patients, so the model misses three in eight of the real cases in women against roughly one in nine in men. A model that leans on sex to push female patients towards a no-disease prediction, then misses more of their true cases, follows the under-diagnosis pattern reported in cardiac prediction (Straw, Rees & Nachev, 2024). This describes how the model uses sex, not a clinical cause, and the female figures rest on only 35 test patients, best read as suggestive rather than exact.

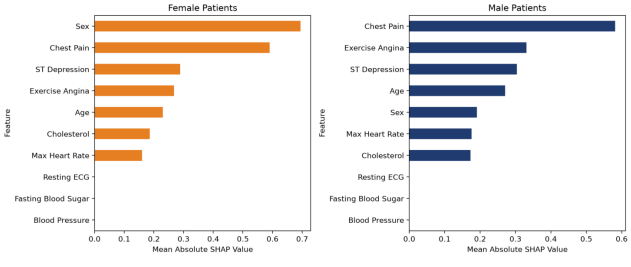


Fig. 2. Mean absolute SHAP value for each feature on the UCI data, computed separately for the two sexes.

C. *Fairness under a narrow base-rate gap*

The Kaggle data sits at the other extreme, with a base-rate gap of less than one percentage point, and the fairness metrics stay close to equality. On the tuned XGBoost none of the four decision-based metrics shows a substantial gap (Table 3): at 0.9792 the disparate impact ratio is well above the 0.8 threshold, so women and men are flagged at almost the same rate. This holds across all three models, with disparate impact ratios of 0.9743, 0.9787 and 0.9792. Because the two sexes have near-equal base rates, the impossibility constraints have little to bind, and the UCI trade-offs do not appear.

The SHAP rankings show no such split. Computed separately for the two sexes, they are almost identical: systolic blood pressure is the top feature for both, while sex falls near the bottom, ninth of eleven features in each group (Fig. 3). Its mean SHAP value is +0.0136 for women and -0.0224 for men, close enough to zero that sex barely moves a prediction. The false negative rates are close as well, 0.3226 for female patients against 0.3203 for male patients. Sex is not acting as a meaningful signal for either group.

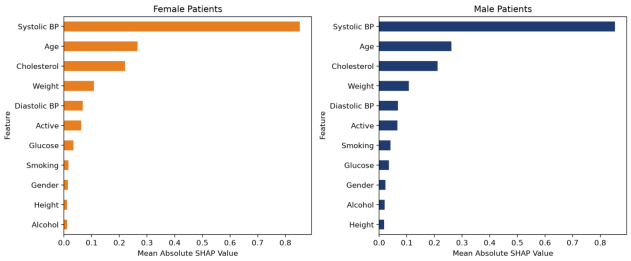


Fig. 3. Mean absolute SHAP value for each feature on the Kaggle data, computed separately for the two sexes.

D. *The effect of mitigation*

Three mitigation methods were then applied to the tuned XGBoost on the UCI data, one at a time: oversampling the minority class with SMOTE-NC, reweighting so that sex and outcome become independent, and optimising a separate decision threshold for each sex. This means each method's effect can be read on its own. Table 5 sets the four fairness metrics, along with accuracy, against the baseline. Every method narrows some of the gaps, but none closes them all. The clearest sign is the predictive parity difference, which does not fall with the others but is higher under every method than the 0.2050 baseline, reaching 0.3316 under threshold optimisation, even as demographic parity, equalised odds and the disparate impact ratio move towards their fair values. Improving one metric therefore worsens another, and none of the three methods avoids this: the choice is not which model is correct but which metric to favour.

TABLE 5. FAIRNESS METRICS AND ACCURACY UNDER EACH MITIGATION METHOD ON THE UCI DATA FOR THE TUNED XGBOOST

Fairness Metric	Baseline	SMOTE-NC	Reweighting	Threshold optimisation
Demographic parity difference	0.4426	0.3889	0.2460	0.1804
Equalised odds difference	0.2580	0.2048	0.0559	0.1596
Predictive parity difference	0.2050	0.2228	0.2985	0.3316
Disparate impact ratio	0.3406	0.3702	0.6016	0.7170
Accuracy	0.8152	0.8043	0.7880	0.7880

Reweighting carries a second cost. The weight it assigns to female patients with disease is the largest of the four groups at 2.0935, which pushes the model towards flagging them but also changes what its probabilities mean. Where the baseline model was reasonably calibrated for both sexes, after reweighting the female curve moves away from the diagonal and the model overestimates the risk of disease for women (Fig. 4). The same reweighting applied to the Kaggle data leaves calibration untouched, since there the weights come out close to one and the two sexes already have near-equal base rates. On the UCI data, by contrast, reweighting shifts the calibration for women, the price paid for lowering their false negatives.

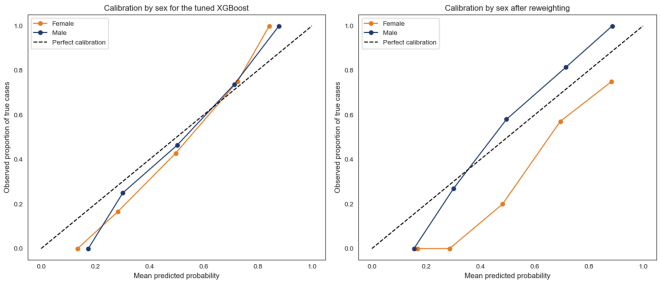


Fig. 4. Calibration by sex on the UCI data, for the tuned XGBoost and after reweighting.

E. *The two datasets compared*

The two datasets were put through the same pipeline: the same models, the same five metrics, the same explanation and mitigation methods. What separates them is the data, though the two sets differ in source and feature composition as well as in base rate. On the UCI data, where the disease rate for men is more than double that for women, the fairness metrics show substantial gaps, sex is prominent in the subgroup explanation, and every mitigation method trades one metric against another. On the Kaggle data, where the two rates sit within a percentage point of each other, the metrics remain near parity, sex falls to the bottom of the ranking, and the same mitigation methods have almost nothing to change.

This is the impossibility result in practice. The theorem holds in both cases, but only where the base rates diverge do the constraints become binding, and only then does improving one metric come at the cost of another. Reading the two together shows how a widening base-rate gap turns a fairness constraint from something satisfied into something that has to be traded, though the datasets differ in more than that gap alone.

F. *The dashboard*

The dashboard turns the same analysis into a fairness trade-off tool for a binary classifier, aimed at the person who trains the model and has to judge, before it reaches a clinician, whether its behaviour is acceptable. Its central design choice is that it does not pick a fair model for that person. Instead, the user sets a tolerance, the largest gap between the two groups they are willing to accept, and the dashboard reports whether the model can meet it. This follows from the impossibility result: since no single setting satisfies every metric once the groups differ in their base rates, the tool presents the conflict rather than resolving it on the user's behalf.

The tolerance defines a feasible region. Working through every pair of thresholds, one for each sex, the dashboard shades the pairs where all four metrics fall within the



Fig. 5. The trade-off the dashboard presents to a model developer: the feasible region at a tolerance of 10%, empty for the UCI data (left) and broad for Kaggle (right). Shaded pairs meet all four metrics; the marker is the default 0.50 threshold.

tolerance. The two datasets then separate cleanly: at a tolerance of 10% the Kaggle region is broad, while the UCI region is empty, and the dashboard states plainly that no setting meets all four metrics (Fig. 5).

An empty region shows that no threshold pair meets all four decision-based metrics at the chosen tolerance, making the practical implications of the impossibility result visible without claiming the theorem alone fixes the shape of the region. The panel reports each both as a value and as a count of patients, so that the demographic parity gap reads as 23 women flagged in every hundred against 67 men on the UCI data (Fig. 6). Reporting the metrics together as patient counts keeps the trade-off from disappearing into one figure and leaves the judgement, as intended, with the user rather than the tool. The full set of dashboard panels, including those described above, is given in Appendix A.

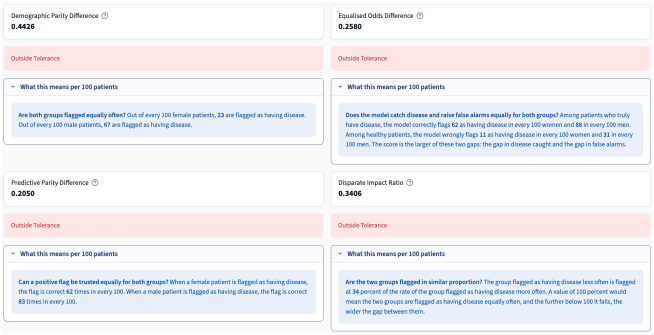


Fig. 6. The same trade-off shown to a model developer: the four fairness metrics for the UCI data, each given both as a value and as a number of patients in every hundred.

V. Discussion / Conclusion

The two datasets were run through the same pipeline and gave sharply different fairness patterns. At the centre of the analysis sits the dashboard, which turns that abstract impossibility into something the user can see and judge. Gender-stratified explanation tied the contrast to how the models used sex: sex is the feature the model leans on most for female patients on the UCI data, yet sits near the bottom of the ranking on the Kaggle data. Read together, the two datasets show that being in the minority is not, on its own, a disadvantage. On the Kaggle data men are the minority at 34.9 per cent of patients, yet the model misses no larger a share of their genuine cases than of the women's. On the UCI data women are the minority and it does miss more of theirs. The disadvantage comes not from the smaller number of women but from the unequal error rates: a false negative rate for women roughly three times that for men, the under-diagnosis long reported for them in cardiac care (Straw, Rees & Nachev, 2024).

The wide base-rate gap leaves a question open: does it reflect a genuine difference in prevalence between the sexes, or the historical under-diagnosis of women? The descriptive checks reported earlier show the gap is not merely an artefact of how the sample was assembled, yet they are descriptive rather than causal, so the data cannot say whether the gap reflects genuine prevalence, historical under-diagnosis, or both, and SHAP shows how the model leans on sex, not why. A causal claim would call for causal fairness, which this work deliberately leaves aside (Kusner et al., 2017); the aim was to locate where the disparity enters, not to prove what causes it. The two datasets also differ in more than their base rates, so the contrast points to that gap without isolating it as a controlled experiment would, and the female figures rest on a test group of only 35 patients, suggestive in direction but not exact in value.

Recording these limits does not weaken the work; documenting them openly is part of what makes a clinical tool honest (Alderman et al., 2025). This fits the stance taken throughout: fairness was not solved but presented, and the dashboard leaves the choice of which metric to give up with the user rather than the tool. The metrics themselves are not read in isolation either, since a fairness gap takes on meaning only alongside the clinical and social context that produces it (Raji et al., 2020). Heart disease was the example here and the dashboard the centre; because the pipeline rests on a binary group label, the same framework could compare two groups defined by another sensitive attribute, such as ethnicity.

VI. Future Work

The dashboard is currently implemented for a binary sensitive attribute, and a natural next step is to extend the analysis from one attribute to several intersecting ones, such as sex and age together. Gender-stratified explanation extends in the same direction: subgroups defined by intersecting attributes rather than one, an approach already seen in recent explainability work (Nnadi et al., 2026). On the mitigation side, in-processing methods, which change the learning algorithm itself, may shape the feasible region differently from the pre- and post-processing methods used here. Beyond these, the work is confined to sex, yet factors with no data in either dataset, such as housing and socioeconomic status, may also contribute to the disparity and could not be examined here.

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APPENDIX

A. Dashboard Panels

Appendix A gives a complete set of views from the fairness trade-off dashboard. The first thirteen figures show the baseline model on both datasets, and the last three show the dashboard under each of the three mitigation methods. The dataset, model and tolerance for each view are given in its caption. The main paper remains self-contained, and these figures are supporting evidence rather than additional results.

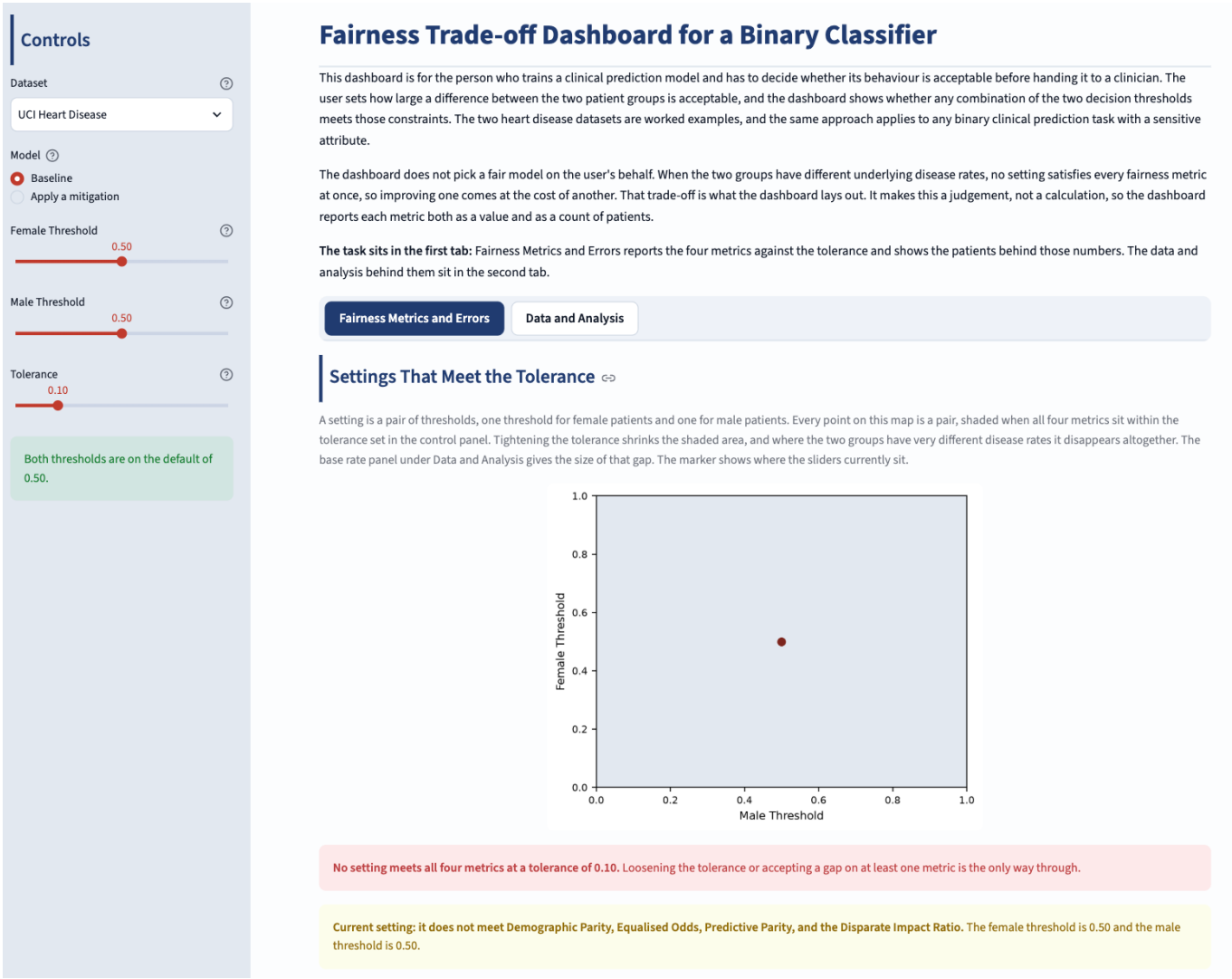


Fig. 7. Settings that meet the tolerance on the UCI data, baseline model at a tolerance of 0.10.

At this tolerance the map is empty: no pair of thresholds keeps all four metrics within the tolerance, and the panel lists the metrics that fail. The marker shows both sliders on the default of 0.50.

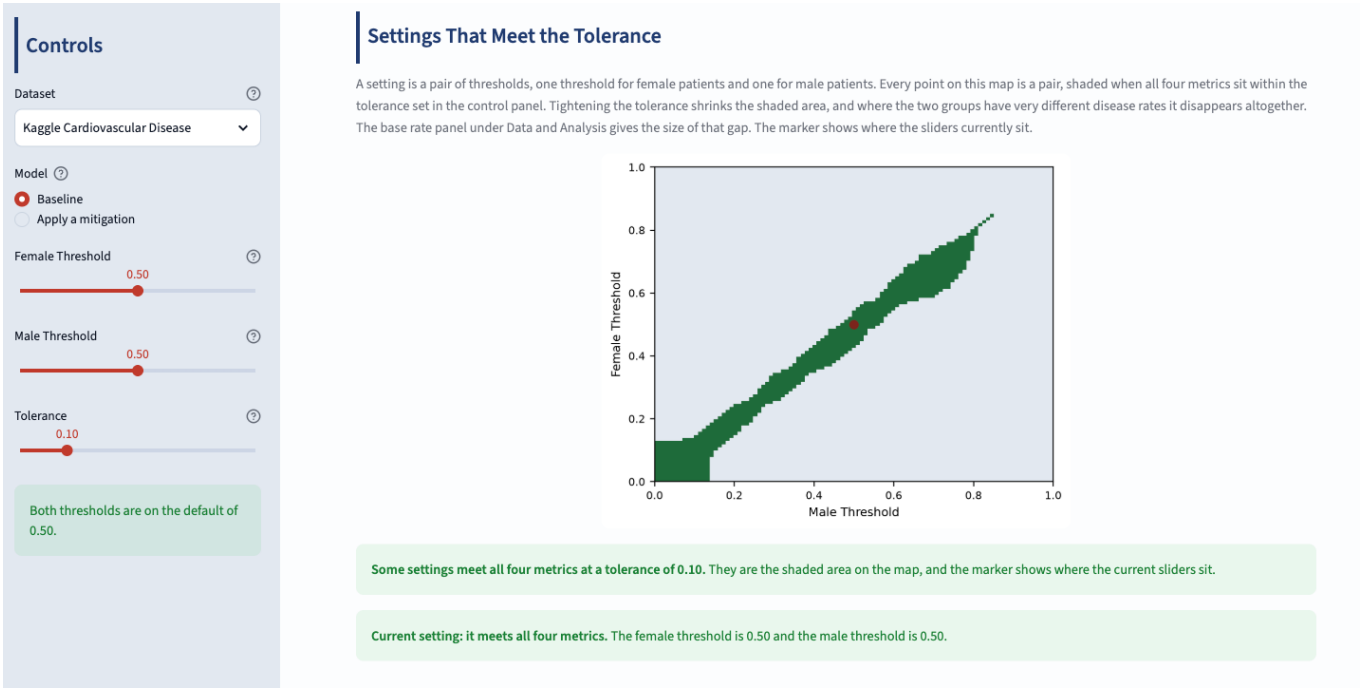


Fig. 8. Settings that meet the tolerance on the Kaggle data, baseline model at a tolerance of 0.10.

Here a broad diagonal band of threshold pairs keeps all four metrics within the same tolerance, and the default setting of 0.50 for both groups falls inside it. The panel confirms that the current pair meets all four metrics.

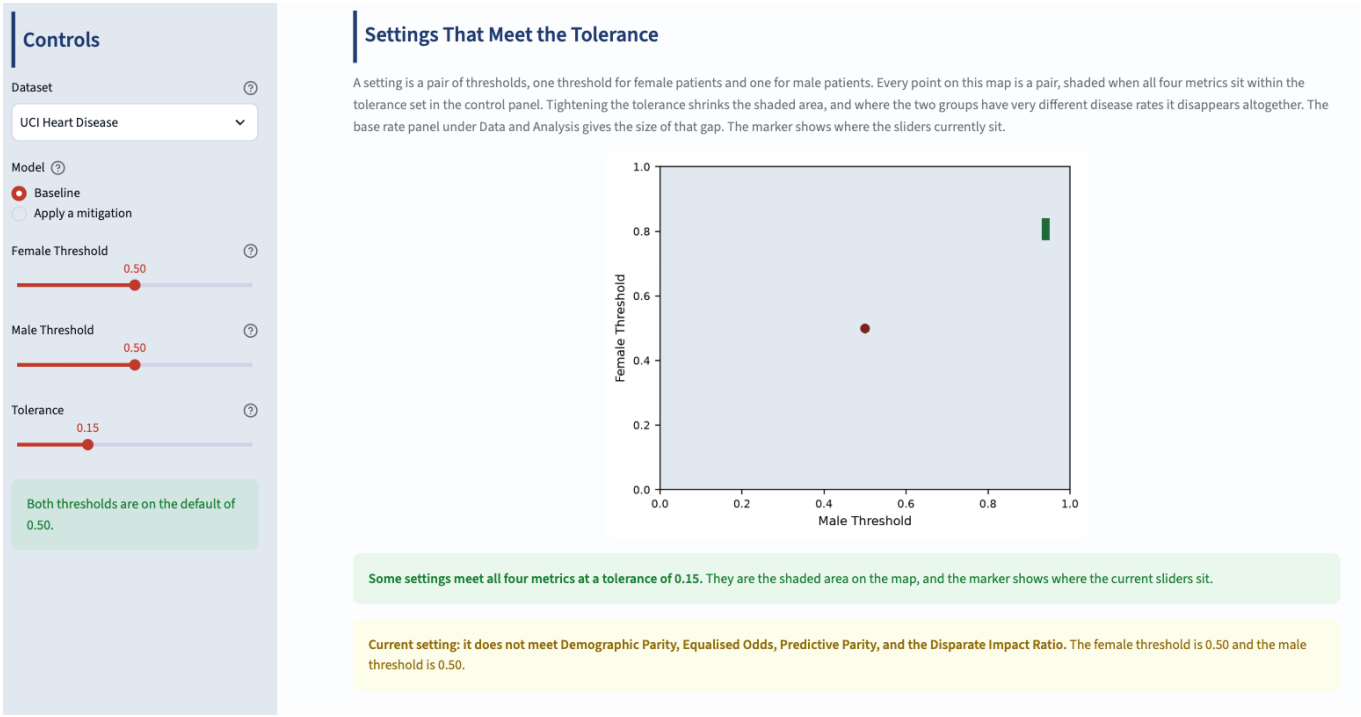


Fig. 9. Settings that meet the tolerance on the UCI data, baseline model at a tolerance of 0.15.

Loosening the tolerance from 0.10 to 0.15 opens a narrow region in the upper right of the map, where both thresholds are high and the male threshold is higher than the female one. The default pair of 0.50 sits well outside it, and the panel still reports that the current setting meets none of the four metrics.

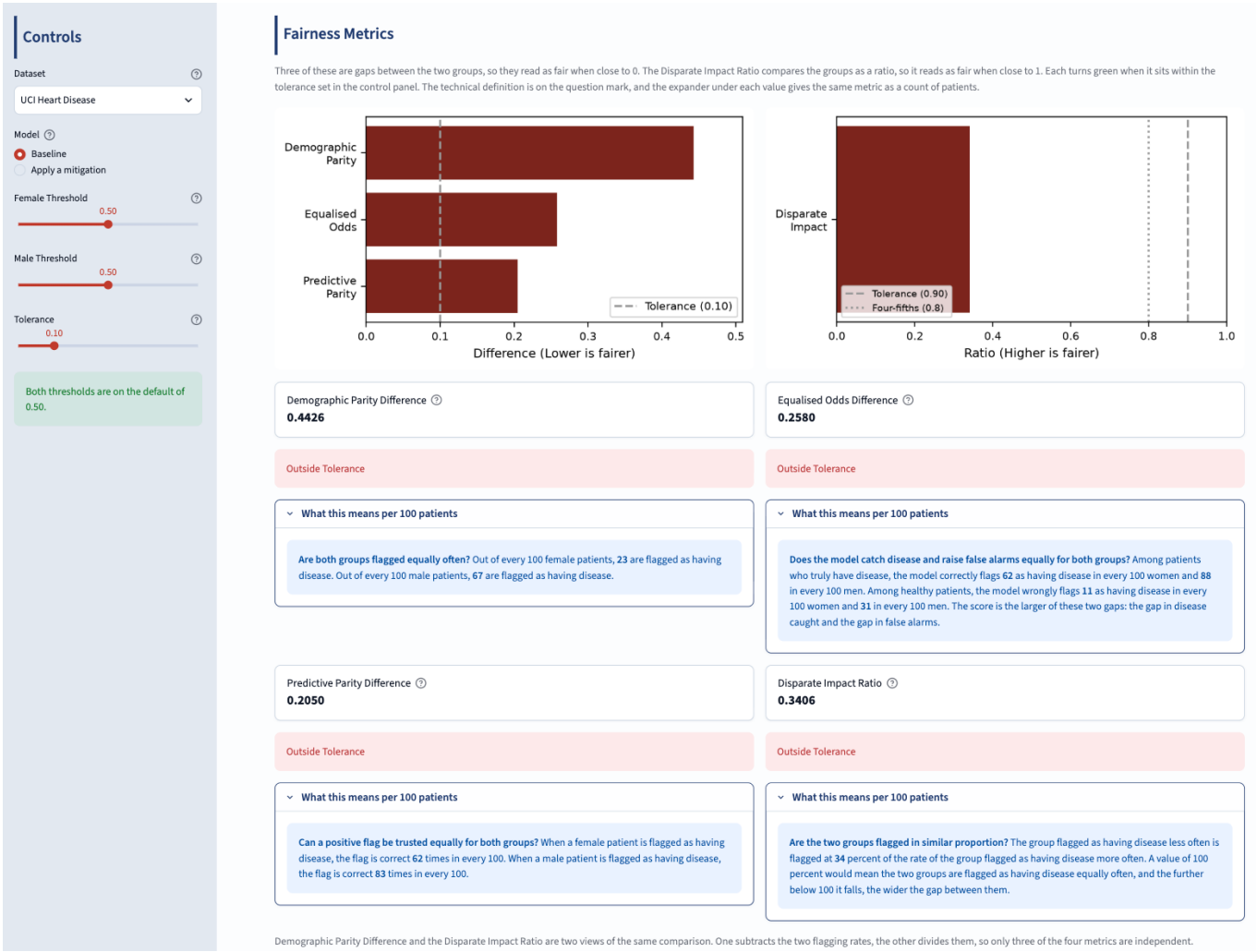


Fig. 10. The four fairness metrics for the UCI data, baseline model at a tolerance of 0.10.

All four metrics fall outside the tolerance, with the demographic parity difference at 0.4426 and the disparate impact ratio at 0.3406, well under the four-fifths reference drawn on the right-hand chart. Each value is restated as a count of patients: 23 women are flagged in every 100 against 67 men, and among patients who truly have disease the model catches 62 women in every 100 against 88 men.

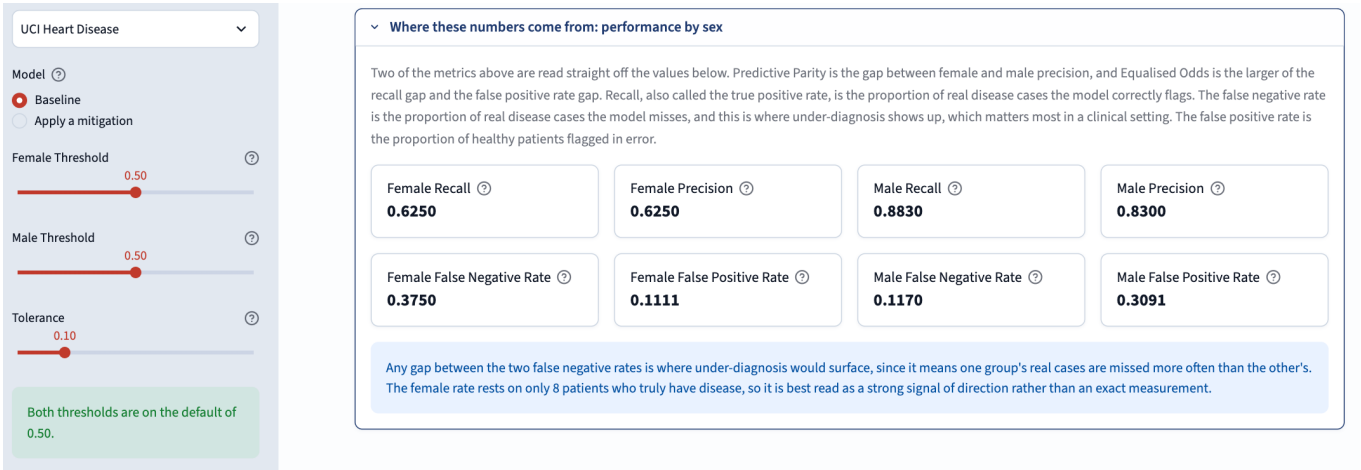


Fig. 11. Performance by sex behind the fairness metrics, UCI data, baseline model.

The panel opens the metrics into the rates they are built from, with recall, precision and the two error rates given separately for each sex. The note beneath records that the female false negative rate rests on 8 patients who truly have disease, so it reads as a direction rather than an exact measurement.

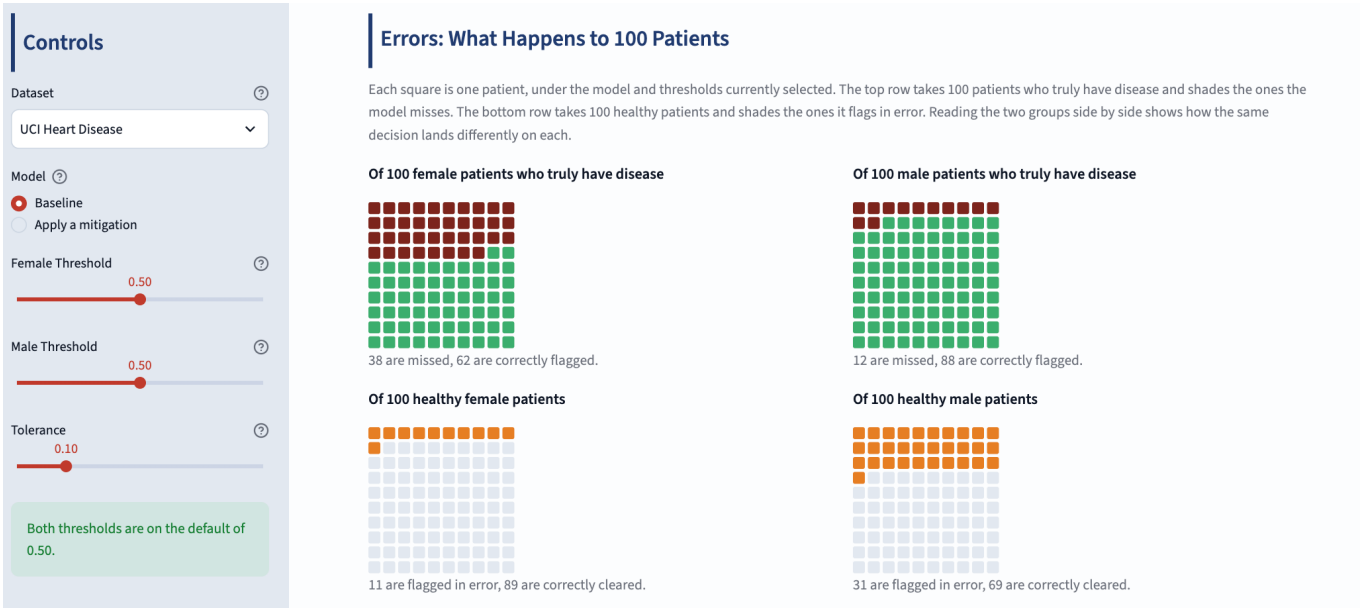


Fig. 12. Errors as counts of 100 patients, UCI data, baseline model at thresholds of 0.50.

Each square is one patient, so the same decision can be read on both groups at once. Of 100 women who truly have disease 38 are missed against 12 of 100 men, while the position reverses among healthy patients, where 11 women are flagged in error against 31 men.

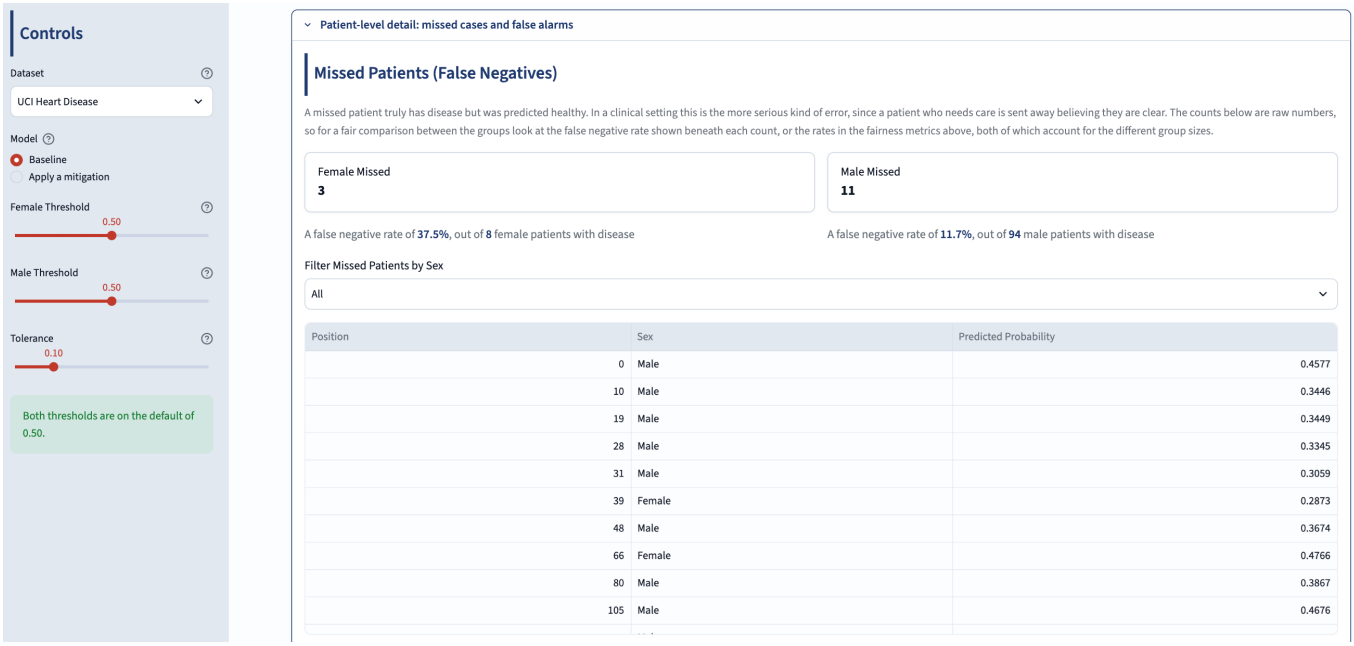


Fig. 13. Missed cases for the UCI data, baseline model at thresholds of 0.50.

The panel lists the individual test patients the model failed to flag, with their position, sex and predicted probability, so a reader can see how close each missed case came to the threshold. The counts are given as raw numbers alongside the rate, since the two groups differ in size.

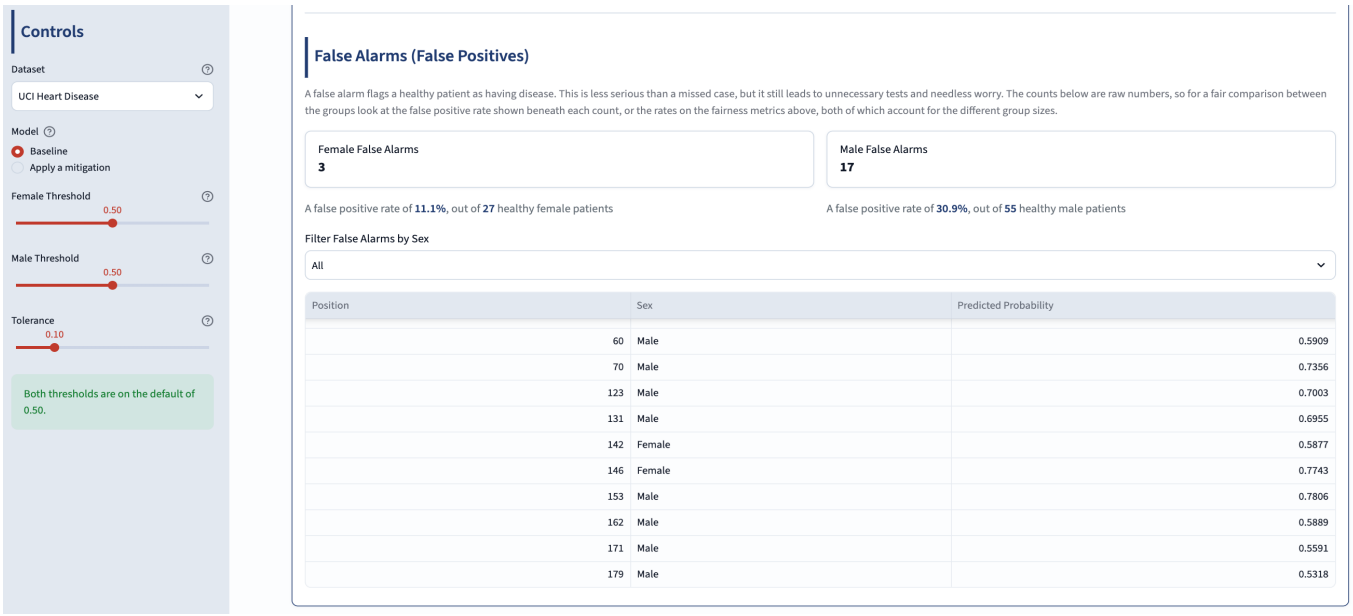


Fig. 14. False alarms for the UCI data, baseline model at thresholds of 0.50.

The same listing is given for healthy patients flagged in error. Taken with the missed cases, the two tables let the errors be traced back to named rows in the test set rather than read only as rates.

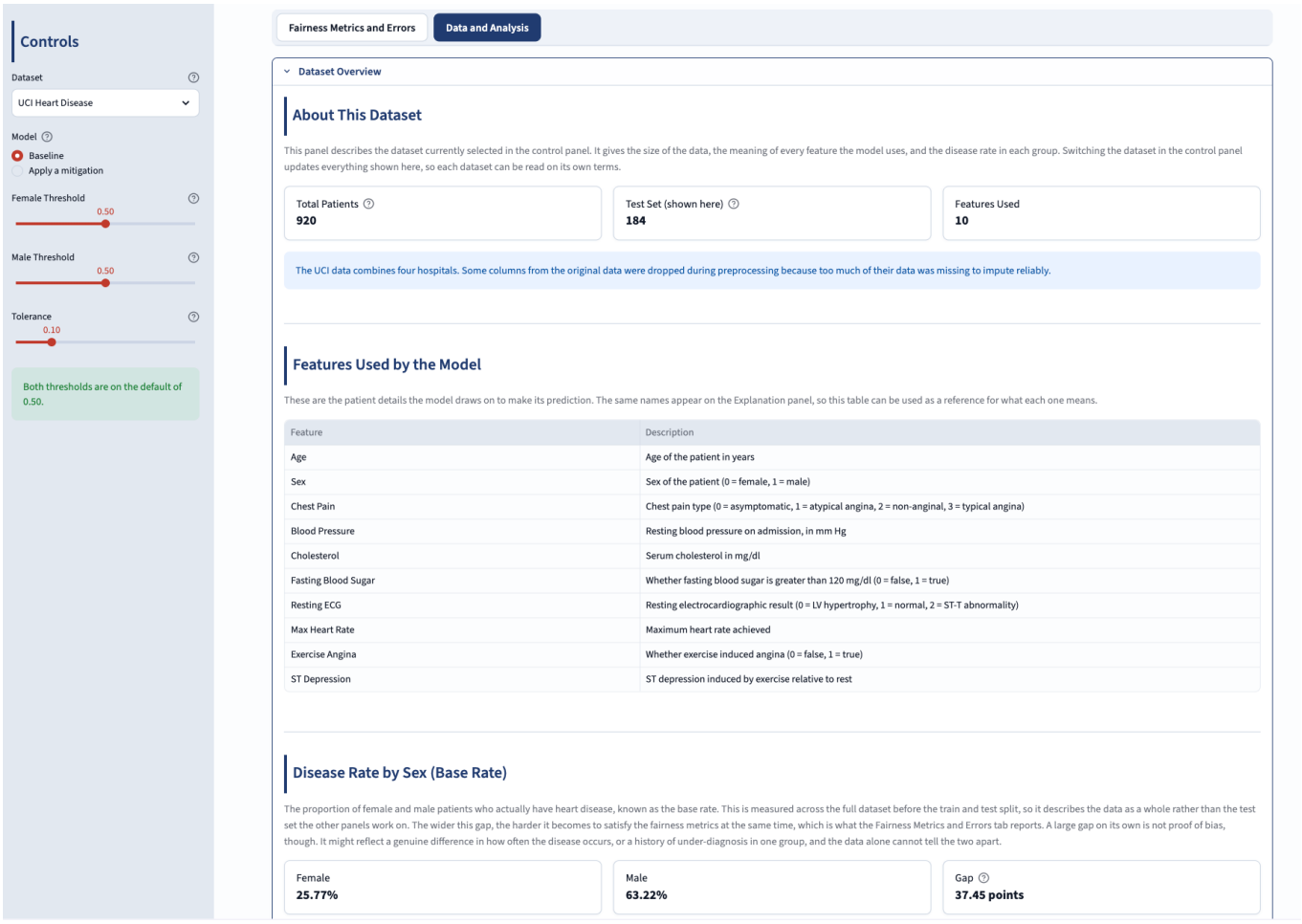


Fig. 15. Dataset overview for the UCI data.

The panel describes the dataset currently selected: the number of patients, the size of the test set, a description of every feature the model uses, and the disease rate in each group. The note beneath the base rates records that a gap on its own is not proof of bias, since the data cannot separate a genuine difference in occurrence from a history of under-diagnosis.

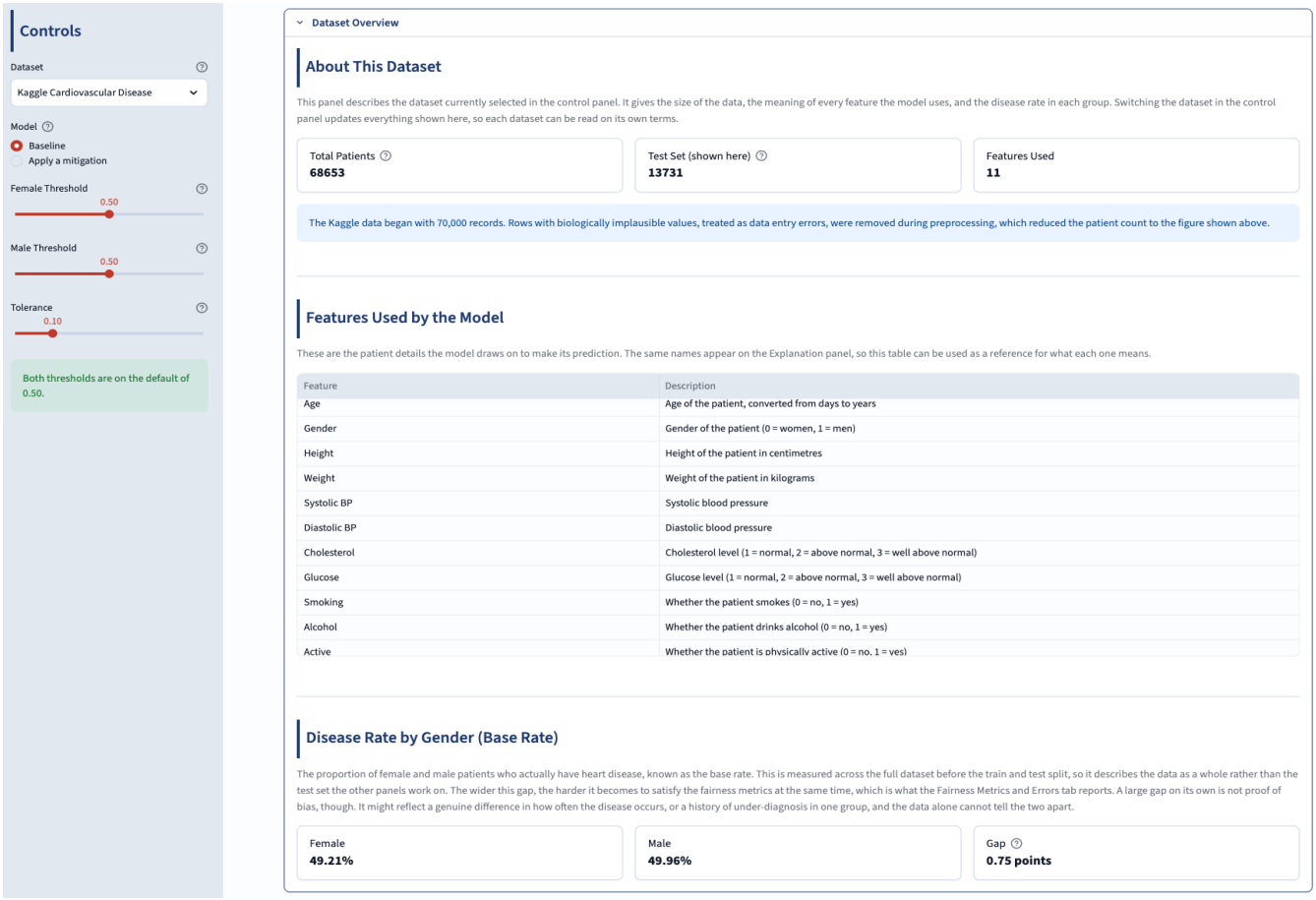


Fig. 16. Dataset overview for the Kaggle data.

Switching the dataset in the control panel updates the whole panel, so the second dataset can be read on its own terms. The feature table and the base rates change with it, and the note above records that rows with biologically implausible values were removed in preprocessing.

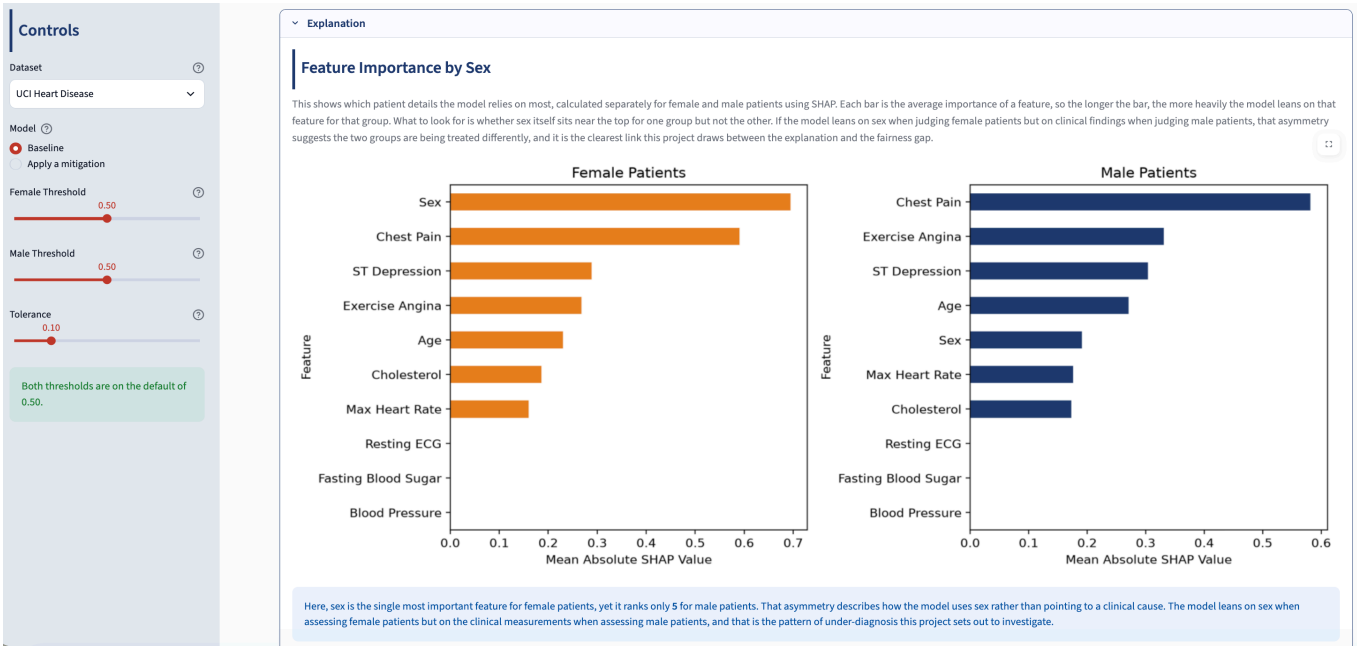


Fig. 17. Feature importance by sex for the UCI data, baseline model.

The panel computes SHAP importance separately for each group, so the two rankings can be read side by side. Sex leads the ranking for female patients and drops to fifth for male patients, and the note beneath records that this asymmetry describes how the model uses sex rather than pointing to a clinical cause.

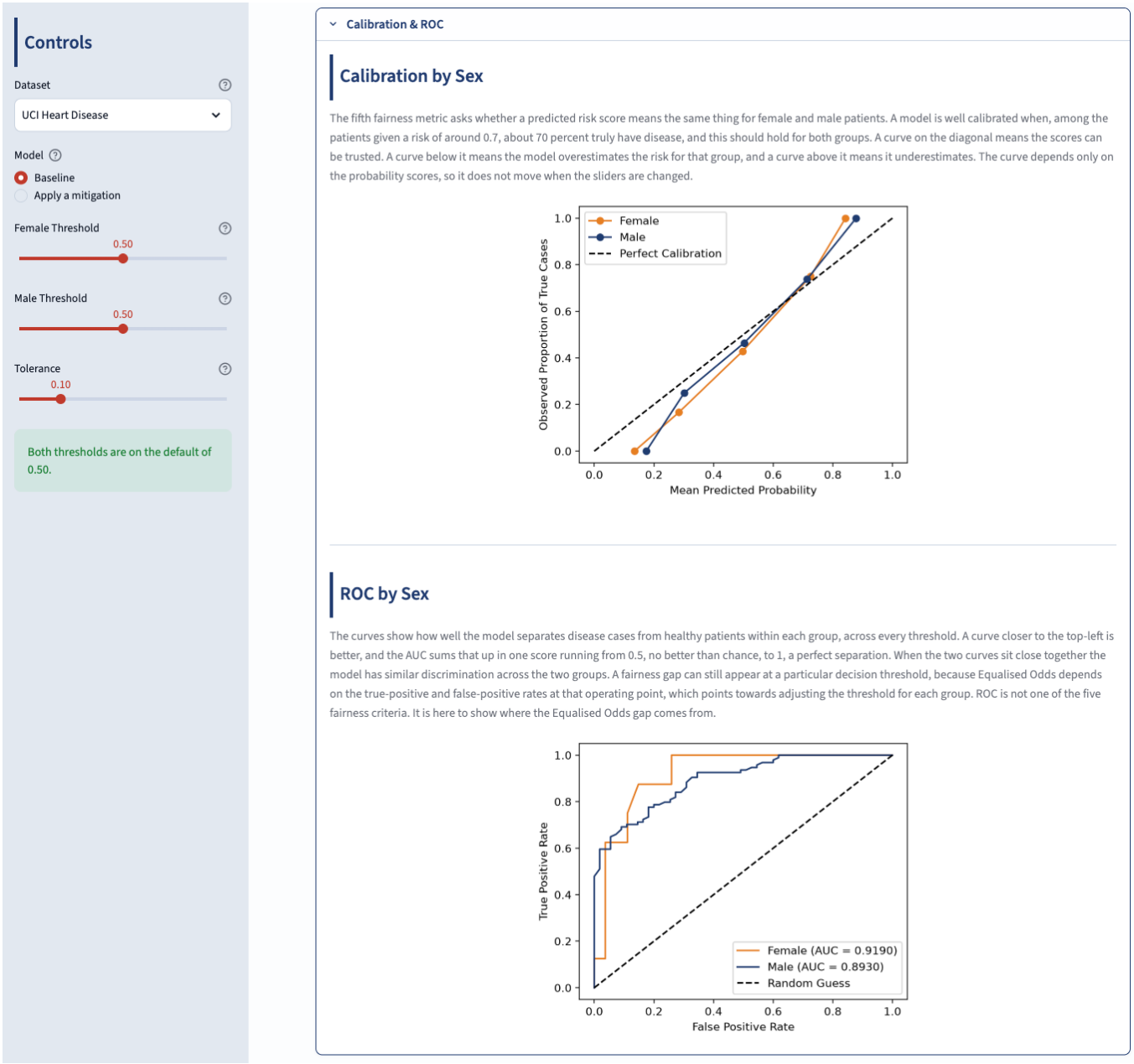
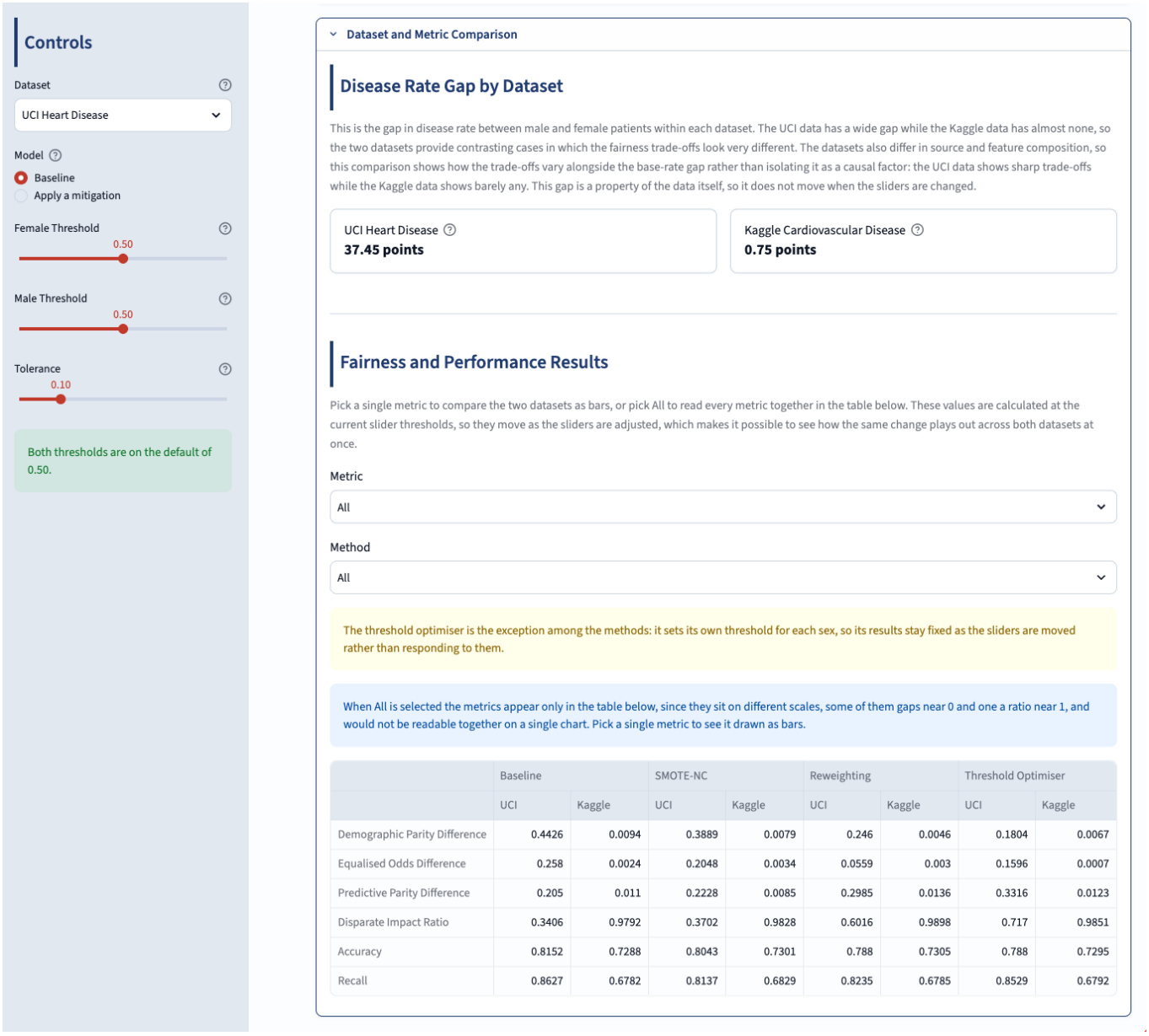


Fig. 18. Calibration and ROC by sex for the UCI data, baseline model.

The two curves sit close together and a little below the diagonal, so a predicted risk score carries broadly the same meaning for both groups even where the threshold-based metrics show large gaps. The panel notes that calibration does not move when the sliders are changed, since it depends only on the probability scores, and that the ROC curves are shown to locate the source of the equalised odds gap rather than as a fairness criterion.



	Baseline		SMOTE-NC		Reweighting		Threshold Optimiser	
	UCI	Kaggle	UCI	Kaggle	UCI	Kaggle	UCI	Kaggle
Demographic Parity Difference	0.4426	0.0094	0.3889	0.0079	0.246	0.0046	0.1804	0.0067
Equalised Odds Difference	0.258	0.0024	0.2048	0.0034	0.0559	0.003	0.1596	0.0007
Predictive Parity Difference	0.205	0.011	0.2228	0.0085	0.2985	0.0136	0.3316	0.0123
Disparate Impact Ratio	0.3406	0.9792	0.3702	0.9828	0.6016	0.9898	0.717	0.9851
Accuracy	0.8152	0.7288	0.8043	0.7301	0.788	0.7305	0.788	0.7295
Recall	0.8627	0.6782	0.8137	0.6829	0.8235	0.6785	0.8529	0.6792

Fig. 19. Dataset and metric comparison, baseline model at thresholds of 0.50.

The panel places the two datasets side by side, first on the base rate gap and then on every metric under each mitigation method at the current thresholds. Reading across the methods shows the trade-off on the UCI data, where an improvement on one metric is paid for on another, while the Kaggle values stay near their fair levels throughout.

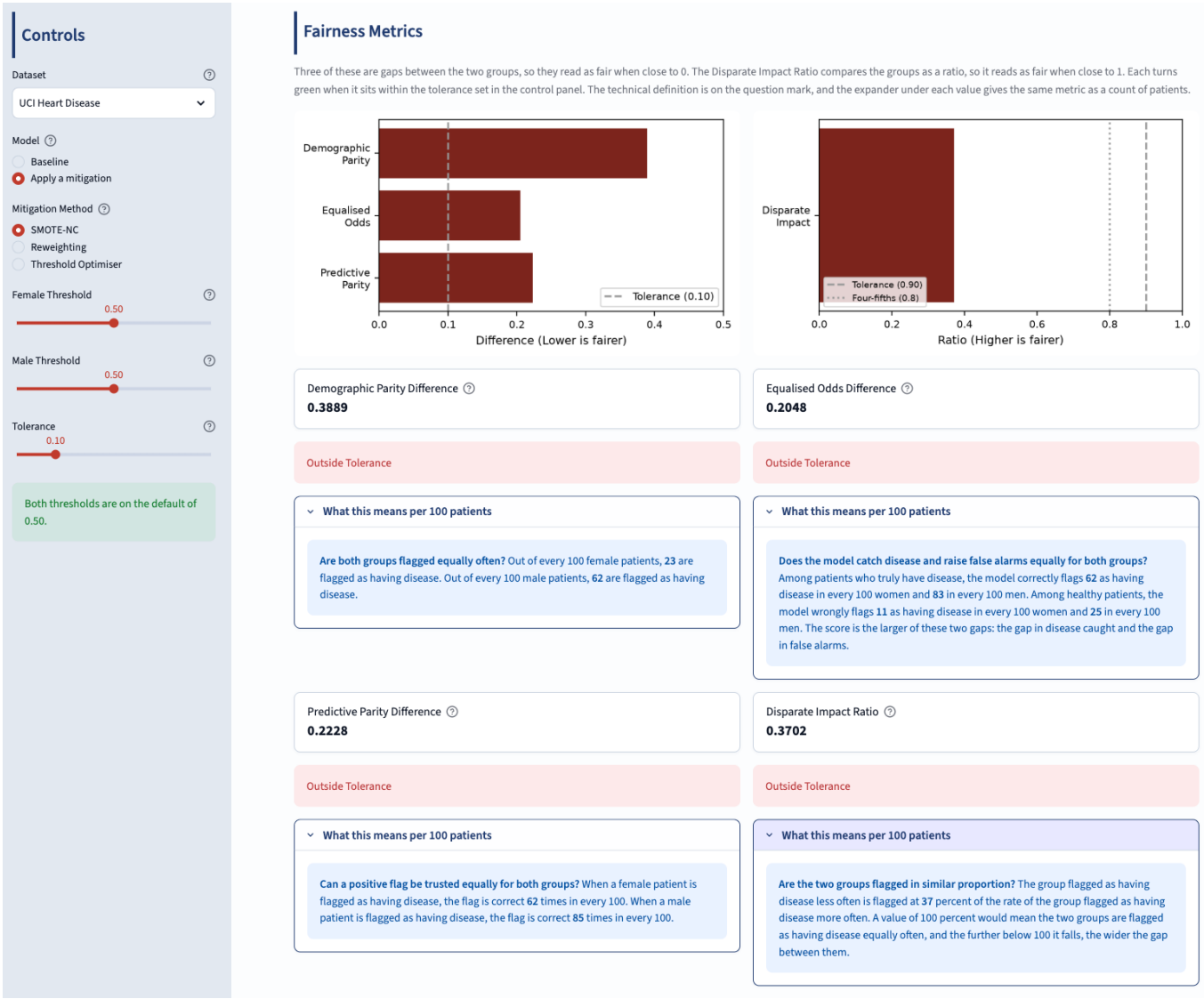


Fig. 20. Fairness metrics after SMOTE-NC, UCI data at thresholds of 0.50.

Selecting a mitigation method in the control panel recomputes the whole tab, so the same four metrics can be read under the resampled model. All four remain outside the tolerance, and the plain-language counts update with them.

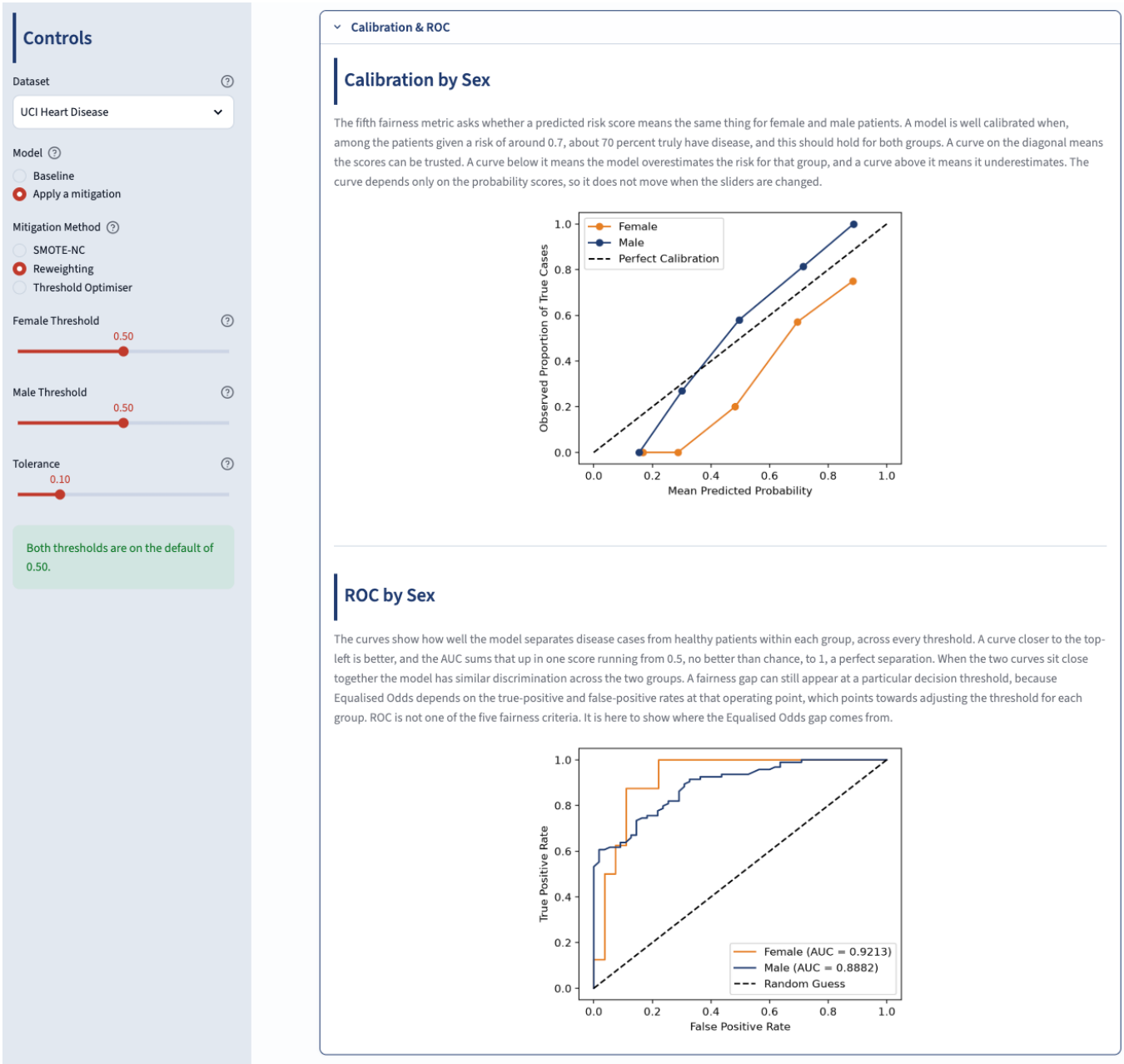


Fig. 21. Calibration and ROC by sex after reweighting, UCI data.

Under reweighting the two curves separate: the female curve falls below the diagonal across most of the range while the male curve sits above it, so the model now overestimates risk for one group and underestimates it for the other. The ROC curves stay close to their baseline shape, so the change appears in the probability scores rather than in how well the model separates the two classes.

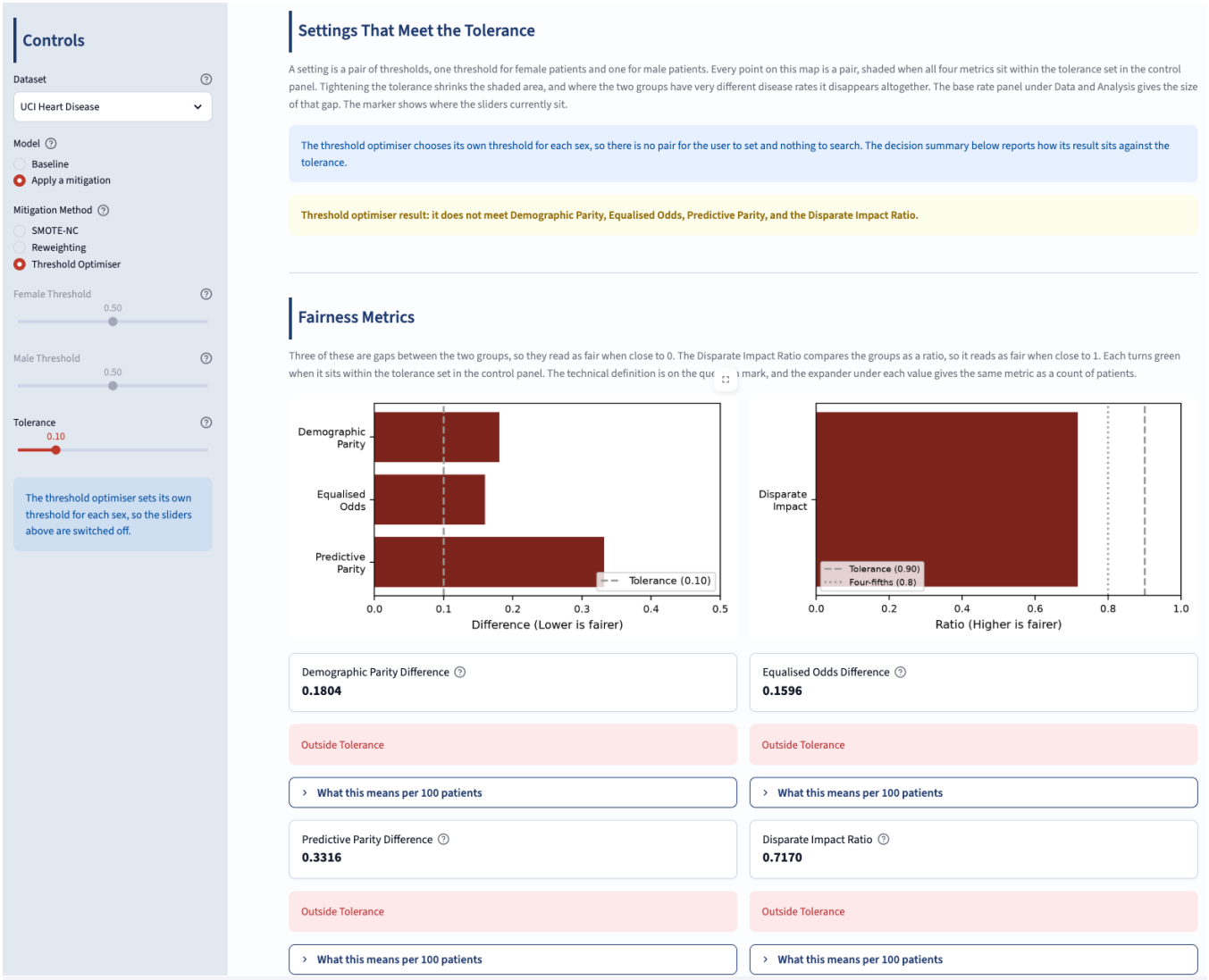


Fig. 22. The dashboard under the threshold optimiser, UCI data.

The optimiser fixes a threshold for each sex, so the sliders are switched off and no feasible region is drawn: there is no pair left for the user to choose. The panel reports the resulting decision against the tolerance instead, and the fairness metrics are shown for it in the same layout as the other methods.

B. Additional Explainability Figures

Appendix B gives the explanation figures referred to in the methodology, taken from the two notebooks. For each dataset the global SHAP ranking is shown first, then the local explanation for two individual patients, and then the same two patients explained with LIME as an independent check on the direction. The gender-stratified comparison itself is in the main paper.

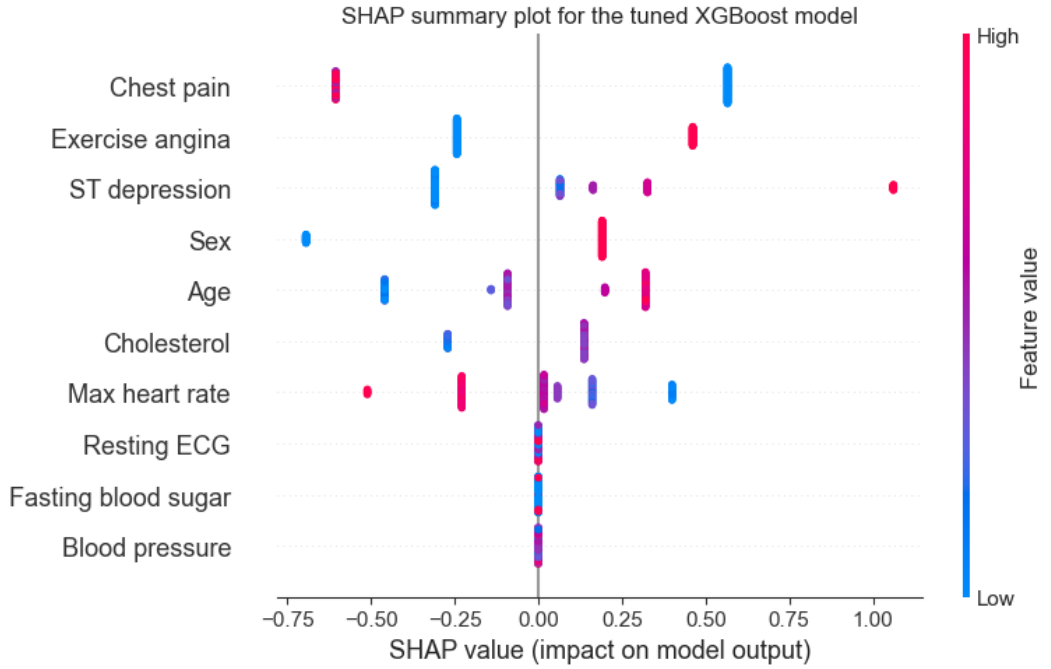


Fig. 23. Global SHAP summary plot for the UCI data, tuned XGBoost.

Each point is one patient, the colour shows whether the feature value was high or low, and the horizontal position shows which way that value pushed the prediction: to the left is towards no disease, to the right is towards disease. Chest pain carries the widest spread, and the blue and red points sitting on opposite sides of zero for sex show the feature pushing the two groups in opposite directions.

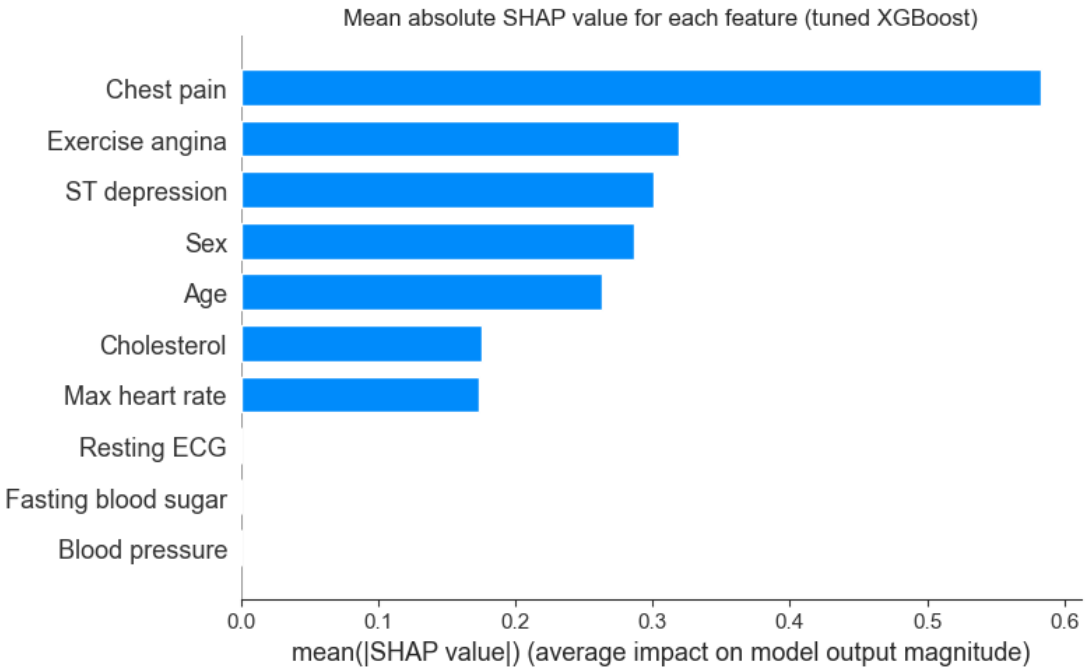


Fig. 24. Mean absolute SHAP value for each feature on the UCI data, tuned XGBoost.

The same values reduced to a single ranking, with the direction dropped. This ranking is computed over all patients together, and the same calculation carried out separately for female and male patients is given in the main paper.

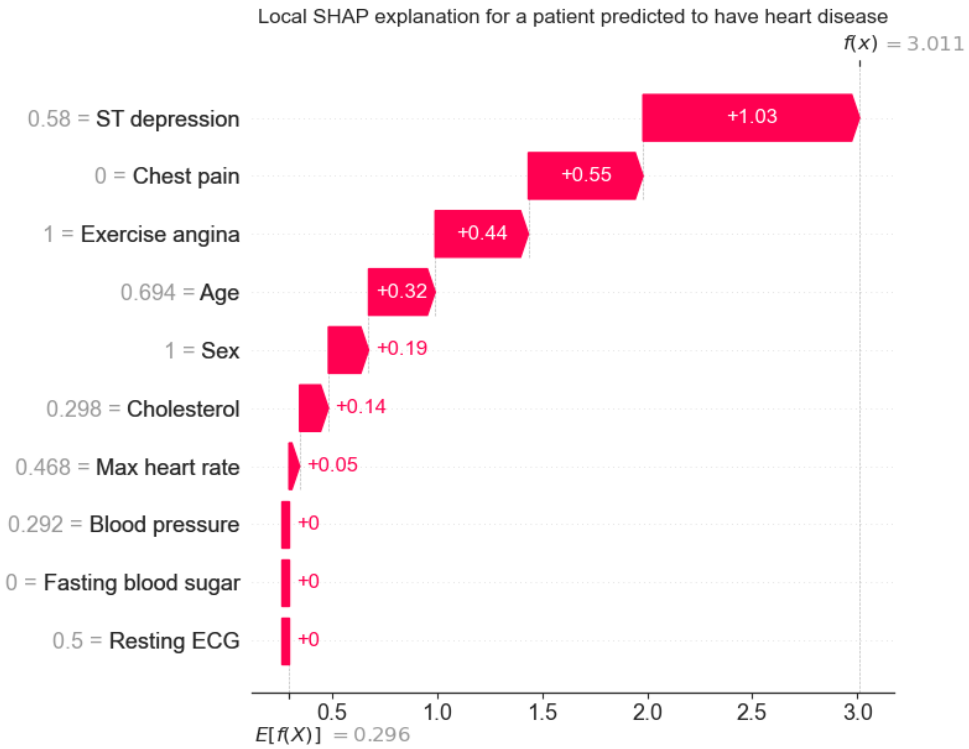


Fig. 25. Local SHAP explanation for a UCI patient predicted to have heart disease.

The plot starts from the base value, the average prediction across the training patients, and adds each feature's contribution until it reaches the output for this patient. Every feature pushes towards disease here, with ST depression the strongest, and sex adds a small push in the same direction for this male patient.

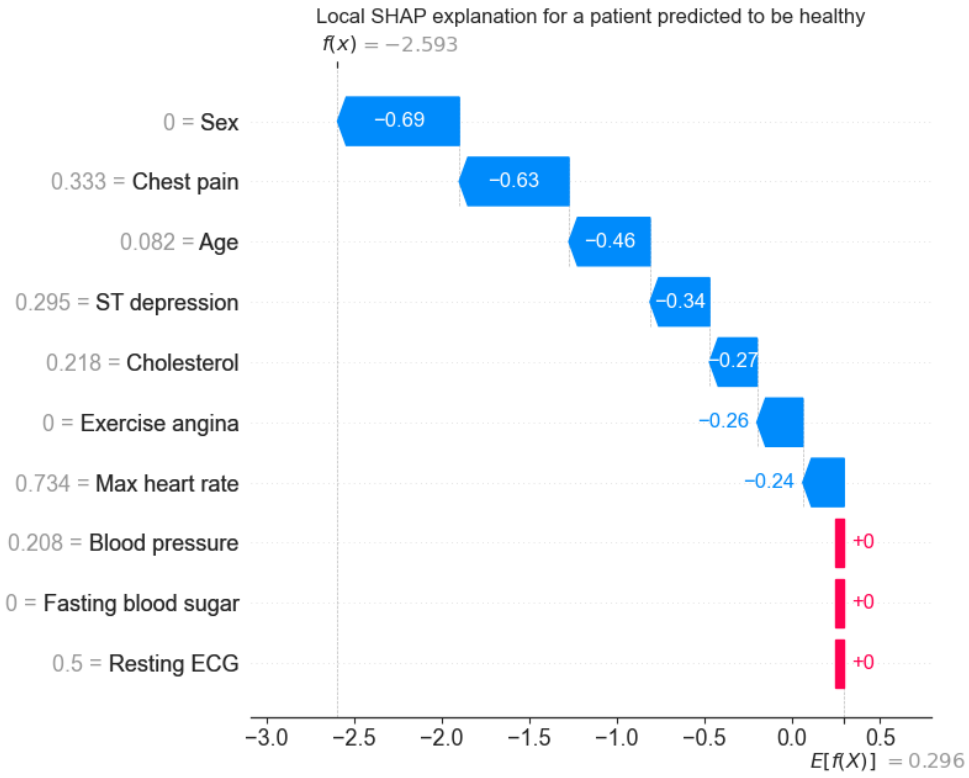


Fig. 26. Local SHAP explanation for a UCI patient predicted to be healthy.

The same construction for a patient predicted to be healthy, where the features push the other way. Sex carries the single strongest push away from disease for this female patient, so the feature acts in opposite directions across the two cases.

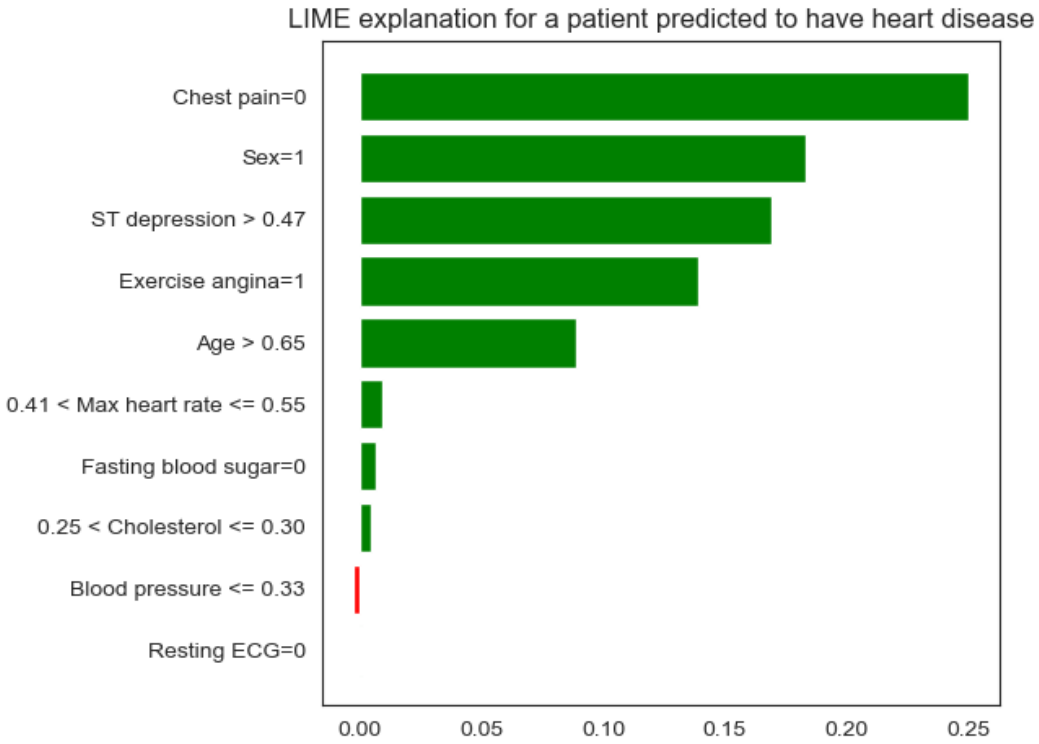


Fig. 27. LIME explanation for the same UCI patient predicted to have heart disease.

LIME perturbs the patient's feature values and fits a simple linear model around that one case, so it checks the SHAP explanation with a different method. Bars to the right push towards disease and bars to the left away from it, and the leading features agree with Fig. 25 even though the order differs.

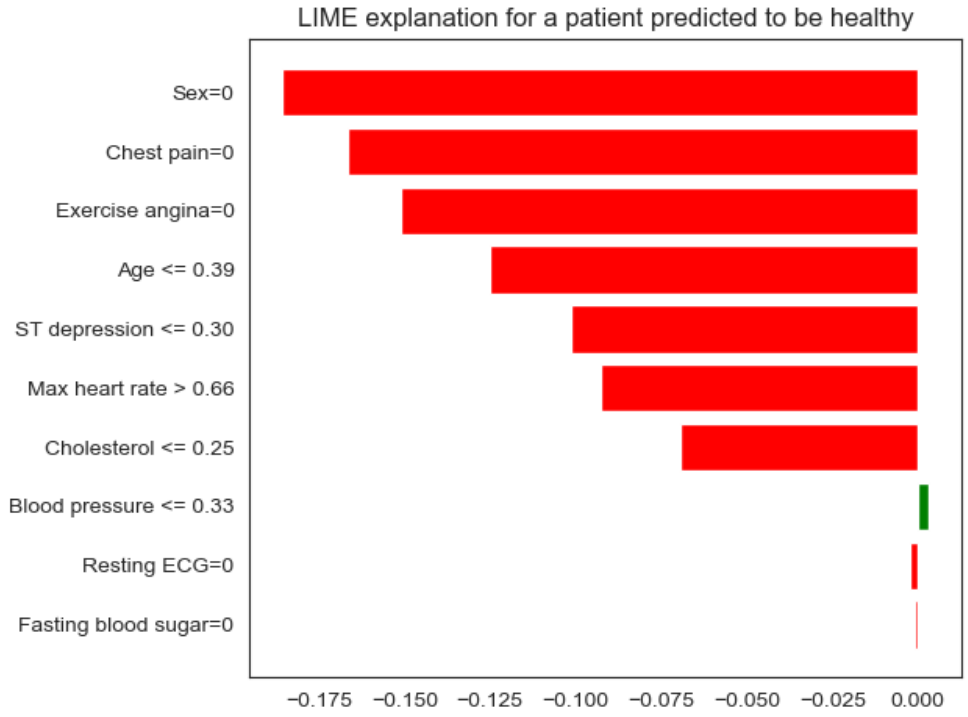


Fig. 28. LIME explanation for the same UCI patient predicted to be healthy.

Here the bars run the other way, away from disease. Sex is the strongest contribution for this female patient under LIME just as it is under SHAP in Fig. 26, so the finding holds under both explanation tools rather than resting on one of them.

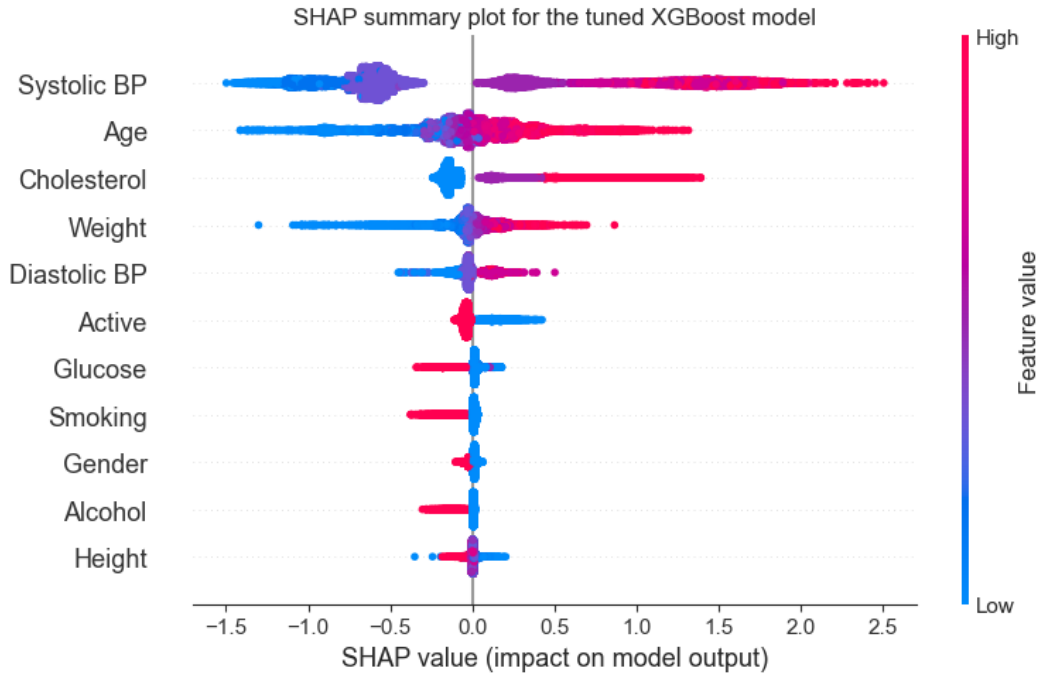


Fig. 29. Global SHAP summary plot for the Kaggle data, tuned XGBoost.

Systolic blood pressure carries by far the widest spread, with age and cholesterol behind it. Gender sits ninth of the eleven features and its points cluster tightly around zero for both groups, so the model draws little on it.

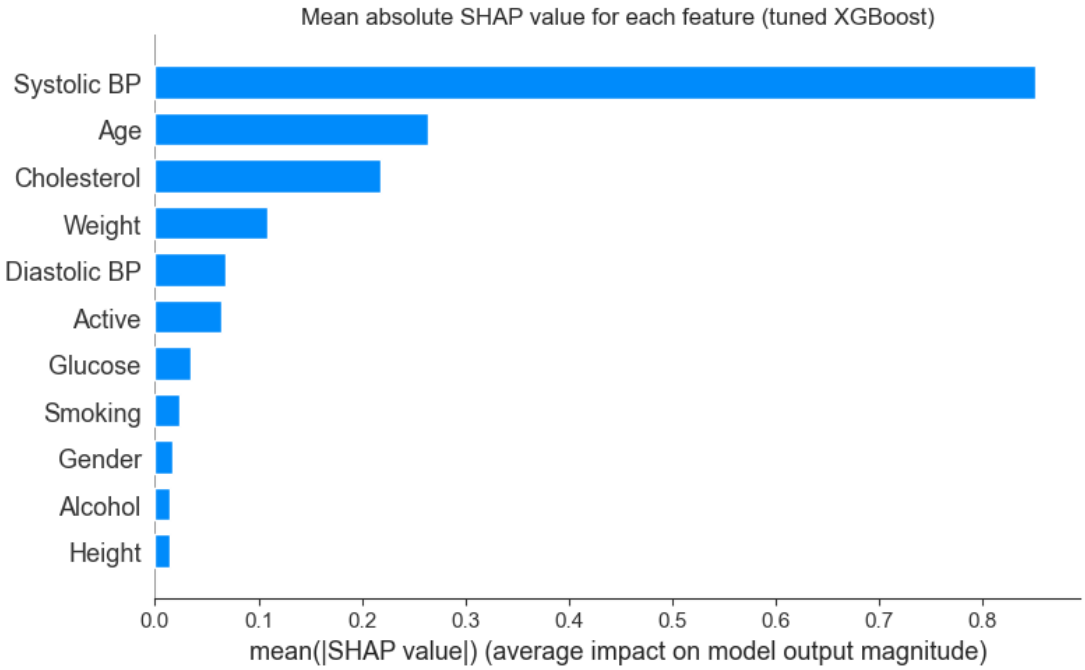


Fig. 30. Mean absolute SHAP value for each feature on the Kaggle data, tuned XGBoost.

The same values reduced to a single ranking, with the direction dropped. Gender is among the three shortest bars, alongside alcohol intake and height, and the gender-stratified version of this calculation is given in the main paper.

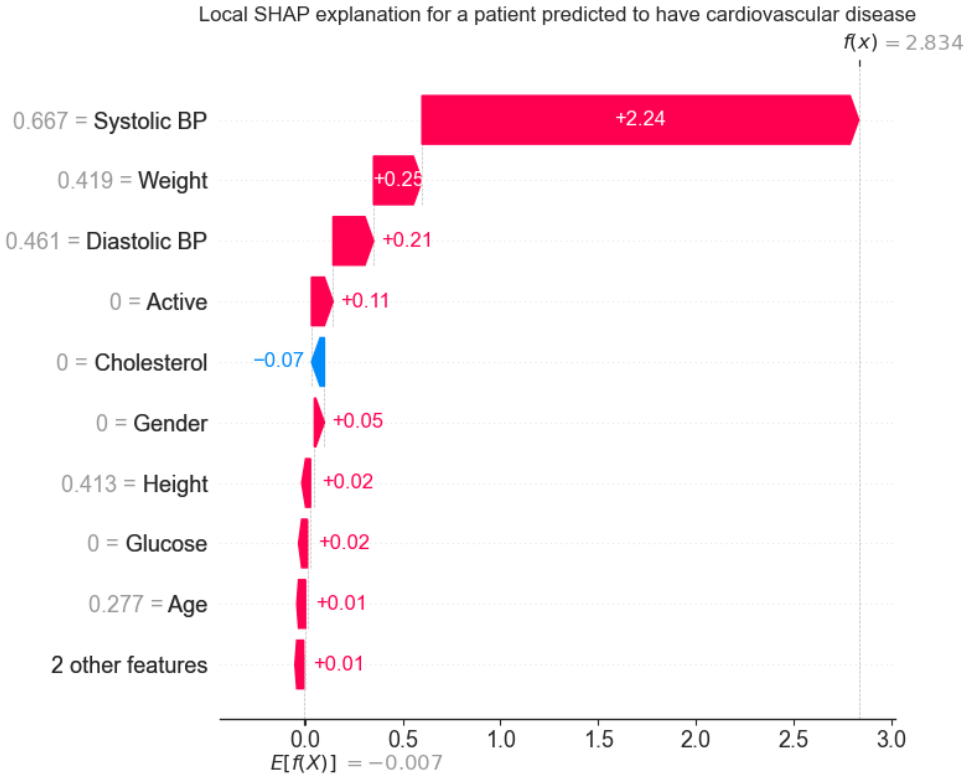


Fig. 31. Local SHAP explanation for a Kaggle patient predicted to have cardiovascular disease.

The plot starts from the base value and adds each feature's contribution until it reaches the output for this patient. Systolic blood pressure supplies almost the whole push towards disease, and gender contributes very little.

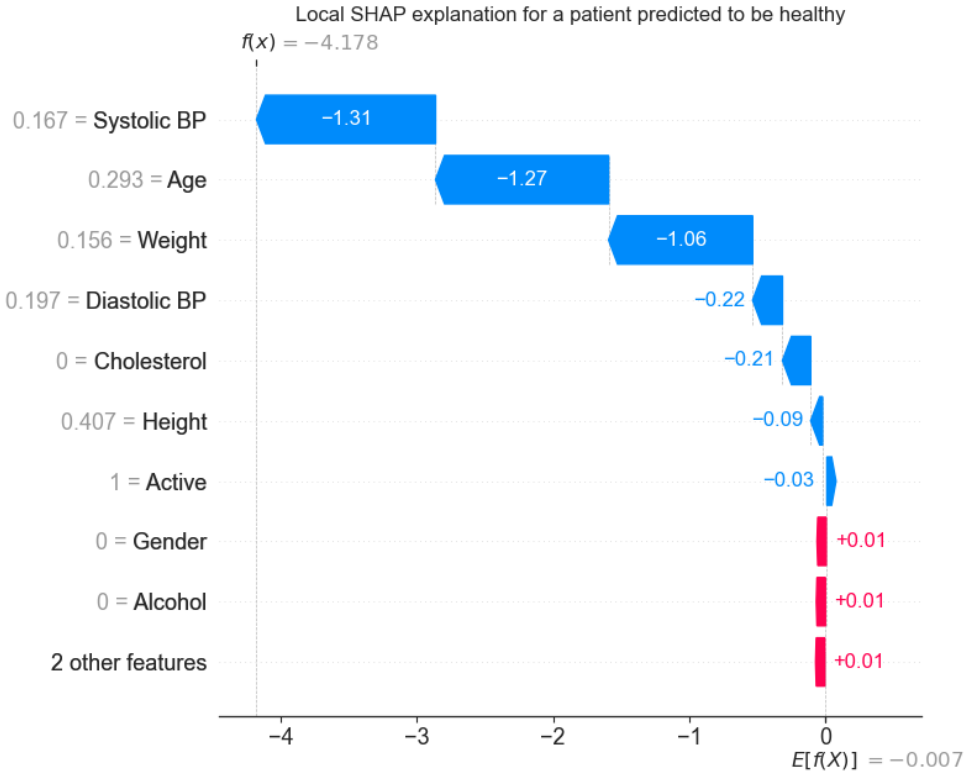


Fig. 32. Local SHAP explanation for a Kaggle patient predicted to be healthy.

The same construction for a patient predicted to be healthy, where the features push the other way. Systolic blood pressure, age and weight carry the movement away from disease, and gender again sits near zero, so the pattern of Fig. 25 and Fig. 26 does not repeat on this dataset.

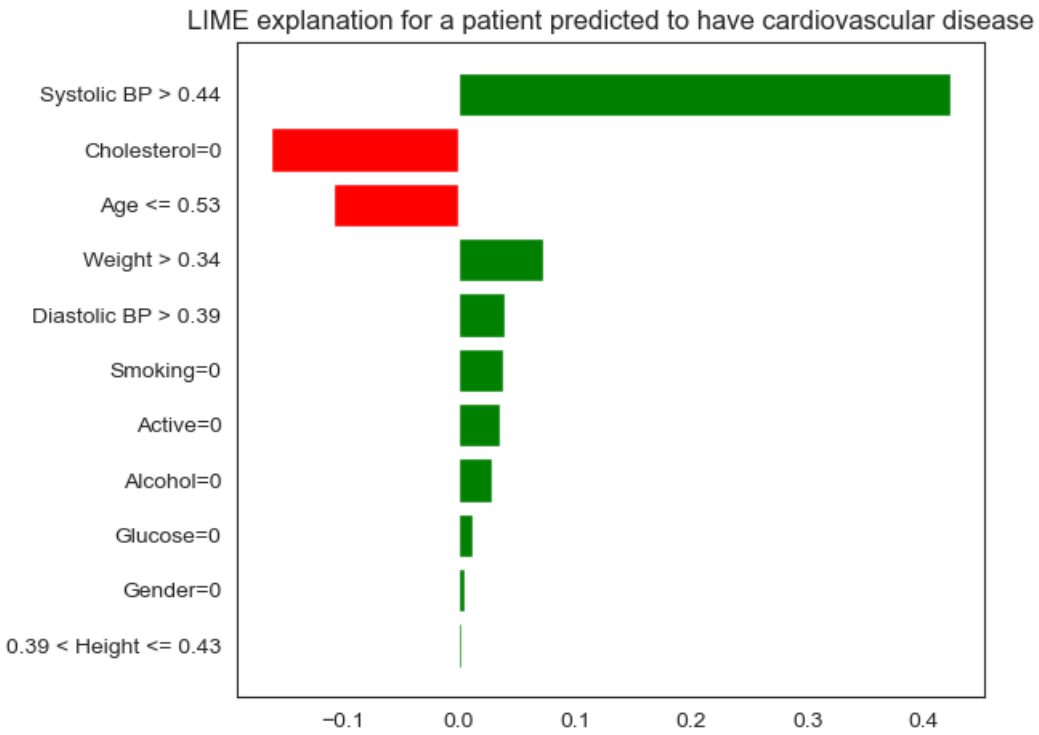


Fig. 33. LIME explanation for the same Kaggle patient predicted to have cardiovascular disease.

LIME checks the SHAP explanation with a different method, fitting a simple linear model around this one case. Bars to the right push towards disease and bars to the left away from it, and systolic blood pressure leads here as it does in Fig. 31.

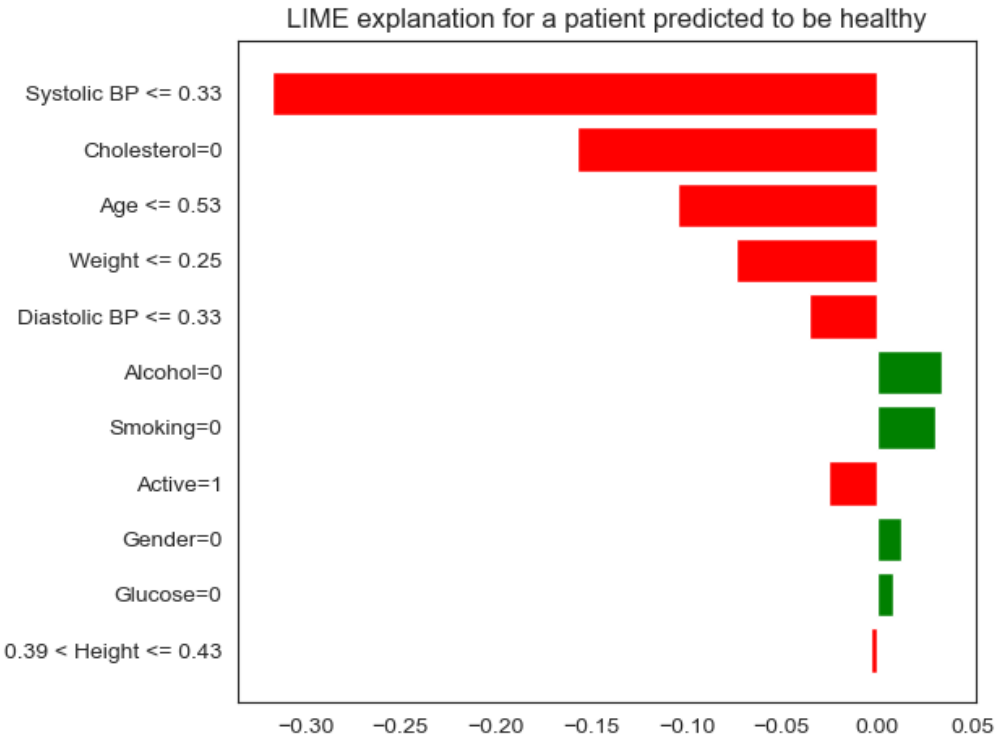


Fig. 34. LIME explanation for the same Kaggle patient predicted to be healthy.

Here the bars run the other way, away from disease. Gender appears near the bottom of the ranking under LIME as it does under SHAP in Fig. 32, so both tools agree that the model barely uses it on this dataset.

C. UCI Data Descriptive Checks

Appendix C gives the checks referred to in the related work, where the base rate gap on the UCI data is examined against two alternative readings: that it follows from the hospital mix, or from an age difference between the two groups. Neither reading accounts for it, which supports treating the gap as a property of the recorded data. These checks are descriptive rather than causal.

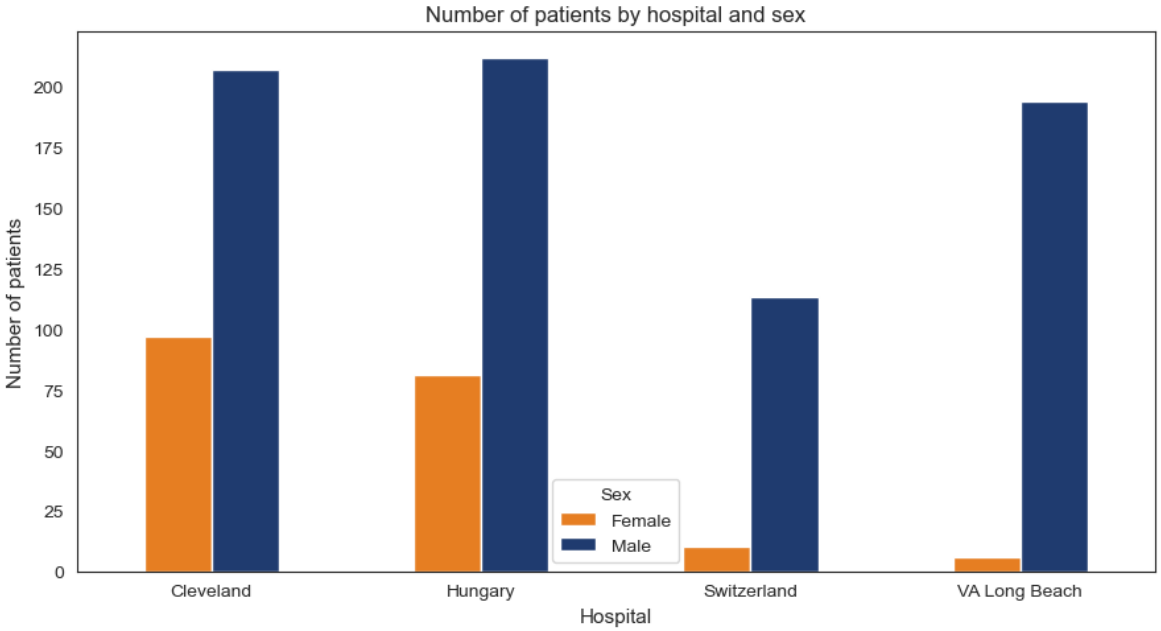


Fig. 35. Number of UCI patients by hospital and sex.

The four hospitals differ sharply in how many female patients they recorded: Cleveland and Hungary hold 97 and 81, while Switzerland and VA Long Beach hold 10 and 6. A rate comparison between the sexes in the next figure is therefore only meaningful for the first two, since the female counts at the other two sites are too small to support one.

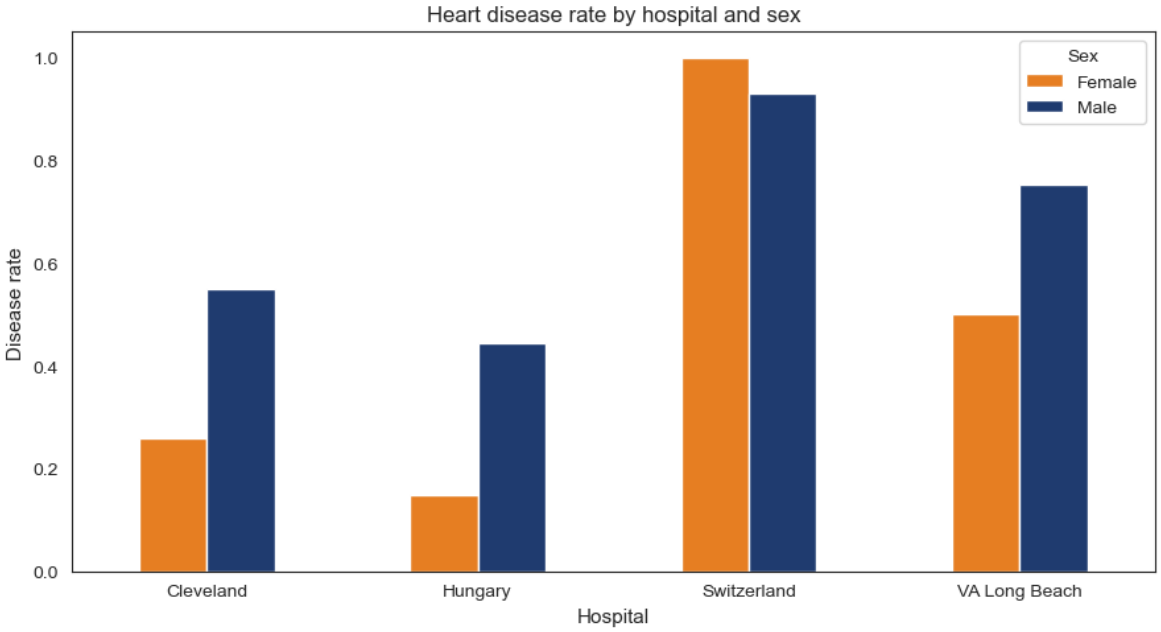


Fig. 36. Heart disease rate among UCI patients by hospital and sex.

Within Cleveland and Hungary, the two hospitals with usable female samples, men are recorded with disease at a higher rate than women. The gap therefore holds inside single hospitals rather than arising only from the mix of sites in the combined dataset.

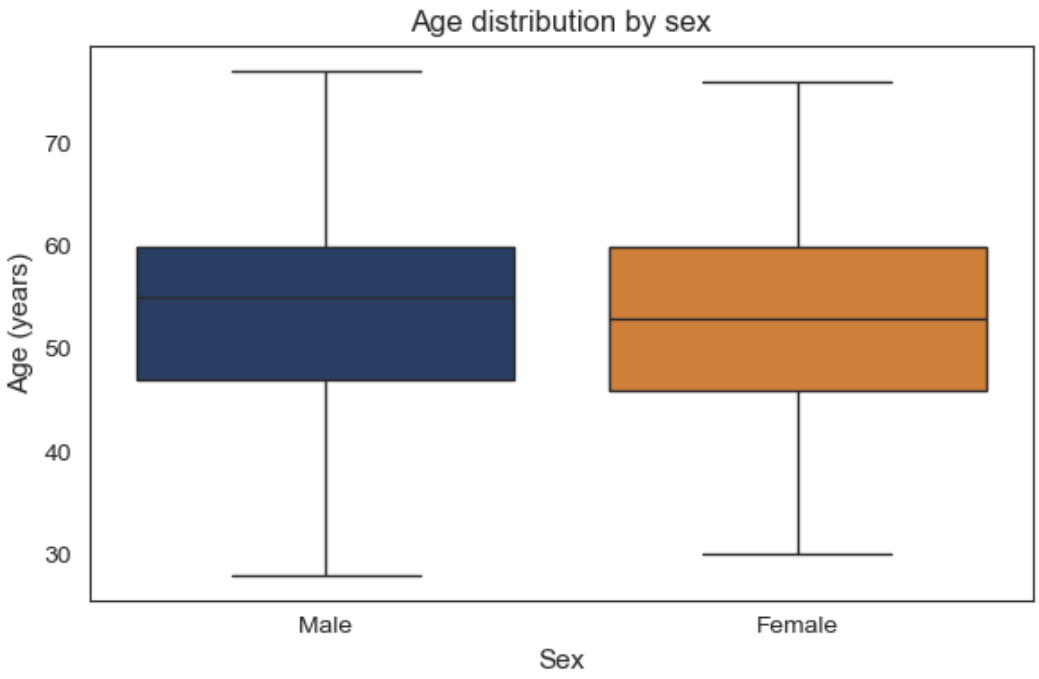


Fig. 37. Age distribution of UCI patients by sex.

The two distributions are close, with mean ages of 52.47 years for female patients and 53.79 for male patients. A difference of about one year is too small to account for the gap in disease rate, so age does not explain it away.